ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the project work entitled *"***ROAD-ACCIDENT ANALYSIS SYSTEM***"* is submitted by

M.TEJO HASINI	**(24B11AI223)**
O.BHEEMESWARI	**(24B11AI305)**
K.SWARNA LATHA	**(24B11AI151)**
P.TAPASYA VALLI	**(24B11AI313)**

in partial fulfilment of the requirements for award of the B.Tech. degree in Artificial Intelligence and Machine Learning for the academic year 2025-2026.

Project Guide

Dr. N.S.L.Kumar Kurumeti, M.Tech., Ph.D
Assistant Professor
Department of AIML

Head of the Department

Dr. M. Venkatesh, M.Tech., Ph.D.
Assoc. Prof.
Department of AIML

DECLARATION

We hereby declare that the Cornerstone project entitled **"ROAD ACCIDENT ANALYSIS SYSTEM"** is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfilment of the requirements for the award of the **B.Tech** degree in **Artificial Intelligence and Machine Learning**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy.

Place : Surampalem

By

M.TEJO HASINI	**(24B11AI223)**
O.BHEEMESWARI	**(24B11AI305)**
K.SWARNA LATHA	**(24B11AI151)**
P.TAPASYA VALLI	**(24B11AI313)**

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ABSTRACT

Road accidents constitute one of the most critical public health and safety challenges of the modern era. Every year, millions of individuals lose their lives or sustain serious injuries due to road accidents across the globe. Understanding the patterns, causes, and contributing factors of road accidents is essential to develop effective preventive strategies and safety measures. The Road Accident Analysis System is designed to address this challenge by employing data mining techniques to analyze and classify road accident data.

This project utilizes the WEKA (Waikato Environment for Knowledge Analysis) data mining tool to apply three powerful classification algorithms: Decision Tree (J48), Naive Bayes, and Random Forest. These algorithms are applied to a dataset consisting of 500 records with attributes including Vehicle Type, Age Group, Gender, Time of Accident, Road Type, and Accident Severity. The accident severity is classified into three categories: Minor, Moderate, and Severe.

The system preprocesses the raw accident data using techniques such as handling missing values, normalization, attribute selection, and data discretization. Following preprocessing, each classification algorithm is applied independently and the results are compared based on accuracy, precision, recall, F-measure, and other performance metrics. The study reveals that the Random Forest algorithm achieves the highest classification accuracy, followed by J48 Decision Tree and Naive Bayes.

The findings of this project provide valuable insights into the major risk factors contributing to road accident severity. The system can assist government agencies, traffic management authorities, and road safety departments in making informed decisions regarding traffic control, road design improvements, and safety campaign targeting. This report presents a detailed account of the system design, implementation, testing, and analysis results.

Keywords: Road Accident Analysis, Data Mining, Machine Learning Classification, WEKA Tool, Random Forest Algorithm, Accident Severity Prediction

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CHAPTER 1: INTRODUCTION

1.1 Introduction of Project Domain

Road accidents represent one of the gravest challenges confronting transportation systems and public health sectors across the world. The World Health Organization (WHO) reports that approximately 1.35 million people die each year as a result of road traffic crashes, making road accidents a leading cause of death globally, particularly among individuals aged 15 to 29 years. In developing countries like India, the problem is especially severe due to rapid urbanization, increased vehicle density, inadequate road infrastructure, and insufficient traffic management systems.

Data mining is the computational process of discovering patterns and knowledge from large datasets. It involves techniques from machine learning, statistics, and database systems, and is widely used in various domains such as healthcare, finance, education, and transportation. In the context of road accident analysis, data mining techniques enable researchers and policymakers to extract meaningful patterns from large collections of accident records, helping to identify risk factors, predict accident severity, and design targeted interventions.

The Road Accident Analysis System developed in this project is a data-driven approach to understanding and categorizing road accidents based on multiple contributing attributes. The system uses the WEKA (Waikato Environment for Knowledge Analysis) tool, which provides an open-source, GUI-based environment for implementing and comparing various machine learning algorithms. WEKA is widely used in academic and research settings for classification, clustering, regression, and association rule mining tasks.

Three well-established classification algorithms are employed in this project: Decision Tree (J48), Naive Bayes, and Random Forest. Each algorithm brings distinct advantages in terms of accuracy, interpretability, and computational efficiency. The dataset contains 500 records of road accident events, capturing attributes such as Vehicle Type, Age Group, Gender, Time of Accident, Road Type, and Accident Severity. The project aims to classify each accident record into one of three severity categories: Minor, Moderate, or Severe.

The motivation behind this project is the urgent need for intelligent, automated systems that can assist traffic authorities, government agencies, and road safety organizations in making evidence-based decisions. By leveraging the power of data mining, this system provides a structured analytical approach to road accident data that goes beyond simple statistical summaries.

1.2 Existing System and Disadvantages

Traditional approaches to road accident analysis have relied heavily on manual data collection and simple statistical methods. Traffic police departments, transport ministries, and research institutions typically gather accident data through field reports, first information reports (FIRs), hospital records, and insurance claims. This data is subsequently compiled into spreadsheets or databases and analyzed using basic statistical tools such as frequency counts, bar charts, and contingency tables.

While these traditional systems have provided valuable insights in the past, they suffer from several significant limitations that hinder comprehensive and accurate analysis of road accident data. The key disadvantages of the existing systems are outlined below.

Manual Data Collection and Processing: The conventional approach to collecting road accident data is largely manual and time-consuming. Field officers fill in paper-based forms that are later entered into digital systems, a process that is prone to human errors, omissions, and inconsistencies. The sheer volume of accident data generated daily makes manual processing inefficient and unreliable.

Inability to Handle Large Volumes of Data: Traditional statistical tools are not designed to handle very large datasets effectively. As accident records accumulate over years, the processing time increases significantly, making real-time or near-real-time analysis impossible using conventional methods.

Limited Predictive Capability: Existing systems primarily focus on describing past events rather than predicting future accident risks. They do not incorporate machine learning models that can classify accident severity or predict outcomes based on input attributes. This limits their utility as proactive decision-support tools.

Lack of Pattern Recognition: Simple statistical analyses fail to detect complex, multi-dimensional patterns in accident data. For instance, the relationship between time of day, road type, vehicle type, and accident severity requires multivariate analysis that standard tools cannot provide.

Delayed Reporting and Decision Making: Due to the manual nature of the existing system, accident reports often take considerable time to compile and analyze. This delay results in slow policy responses and inadequate allocation of resources to high-risk accident zones.

Poor Integration: Existing systems often operate in silos, with data from traffic police, hospitals, and insurance companies stored separately with little or no integration. This fragmented data landscape prevents a holistic view of the accident scenario.

1.3 Proposed System and Advantages

The proposed Road Accident Analysis System is an intelligent, data mining-based solution that overcomes the limitations of traditional accident analysis approaches. The system leverages the WEKA data mining tool to implement and evaluate multiple machine learning classification algorithms, enabling automated, accurate, and insightful analysis of road accident data.

The proposed system takes a dataset of 500 road accident records as input, preprocesses the data to handle missing values and normalize attributes, and then applies three classification algorithms — J48 Decision Tree, Naive Bayes, and Random Forest — to classify accident severity into three categories: Minor, Moderate, and Severe. The system also provides comparative performance evaluation of the three algorithms to identify the most accurate and reliable approach for accident severity prediction.

Advantages of the Proposed System:

Automated Analysis: The system automates the entire data analysis pipeline, from data preprocessing to classification and performance evaluation, significantly reducing the time and effort required compared to manual methods.

High Accuracy: By employing advanced machine learning algorithms such as Random Forest and J48, the system achieves high classification accuracy, providing reliable predictions of accident severity based on input attributes.

Pattern Recognition: The system can detect complex, non-linear patterns in accident data that are invisible to simple statistical analysis. This helps identify hidden risk factors and accident-prone conditions.

Multi-Algorithm Comparison: The system evaluates multiple classification algorithms and compares their performance, enabling selection of the most appropriate model for different use cases and datasets.

Scalability: The system is designed to handle growing datasets, making it suitable for large-scale accident databases maintained by national and state-level transportation authorities.

Decision Support: The insights generated by the system provide actionable intelligence to road safety authorities, enabling targeted interventions such as improving road conditions at high-risk locations and deploying traffic management personnel during peak accident hours.

Open-Source Tool: By using WEKA, the system is built on a freely available, widely supported platform, making it accessible to educational institutions, research organizations, and government departments without additional software licensing costs.

1.4 Objectives of the Project

The primary objective of the Road Accident Analysis System is to apply data mining classification techniques to analyze road accident data and predict accident severity. The specific objectives of the project are as follows:

1. To collect and preprocess a dataset of 500 road accident records containing attributes such as Vehicle Type, Age Group, Gender, Time of Accident, Road Type, and Accident Severity.

2. To apply data preprocessing techniques including missing value handling, attribute normalization, and feature selection to prepare the dataset for classification.

3. To implement and evaluate three classification algorithms — Decision Tree (J48), Naive Bayes, and Random Forest — using the WEKA data mining tool.

4. To classify road accidents into three severity categories: Minor, Moderate, and Severe, based on the input attributes.

5. To compare the performance of the three classification algorithms using evaluation metrics such as accuracy, precision, recall, F-measure, and ROC area.

6. To identify the most significant attributes that contribute to road accident severity through attribute importance analysis.

7. To generate meaningful insights and visualizations that can assist road safety authorities and policymakers in making evidence-based decisions.

8. To demonstrate the effectiveness of data mining techniques in transforming raw accident data into actionable knowledge for improving road safety.

These objectives collectively aim to establish a robust and reliable system for road accident severity analysis that can be extended and adapted for use in real-world traffic management scenarios.

1.5 Organization of Project

The Road Accident Analysis System project is developed using a systematic methodology that involves multiple stages starting from data collection to final prediction of accident severity. The project is organized in a structured manner to ensure accurate analysis using Data Mining techniques and WEKA tool.

Initially, the accident dataset was collected containing important attributes such as Vehicle Type, Age Group, Gender, Time of Accident, Road Type, and Accident Severity. The dataset consists of 500 records which are used for training and testing the classification models.

After collecting the dataset, the data preprocessing step was performed using WEKA tool. In this stage, missing values were removed, inconsistent data was corrected, and categorical values were converted into suitable format required for classification algorithms. Data preprocessing improves the quality of dataset and increases prediction accuracy.

Once preprocessing is completed, classification algorithms are applied to analyze accident severity patterns. The algorithms used in this project are Decision Tree (J48), Naive Bayes, and Random Forest. These algorithms are selected because they provide efficient classification results and are widely used in data mining applications.

The dataset is divided into training data and testing data. The training dataset is used to build classification models, while the testing dataset is used to evaluate model accuracy. WEKA tool provides performance measures such as accuracy, precision, recall, and confusion matrix to analyze classifier performance.

After classification, the results obtained from different algorithms are compared to determine the best performing model. Random Forest generally provides higher accuracy due to combination of multiple decision trees.

Finally, the predicted results are displayed in the form of reports and accuracy outputs. The system helps in identifying accident severity level as Minor, Moderate, or Severe based on input conditions.

Thus, the project is organized in the following sequence:

Data Collection → Data Preprocessing → Classification using J48, Naive Bayes, Random Forest → Result Evaluation → Prediction Output

This structured approach ensures efficient analysis of accident data and helps in improving road safety decision making.

CHAPTER 2: REQUIREMENTS ANALYSIS

2.1 Introduction

Requirements analysis is the foundational phase of any software or data-driven project. It involves the systematic identification, documentation, and validation of the requirements that the system must satisfy in order to achieve its intended objectives. In the context of the Road Accident Analysis System, requirements analysis encompasses the identification of hardware and software needs, the specification of the dataset, and the definition of the functional and non-functional properties of the system.

The requirements for this project are categorized into hardware requirements, software requirements, and data requirements. Hardware requirements describe the minimum and recommended physical infrastructure needed to run the WEKA tool and associated data processing activities. Software requirements specify the operating system, data mining tool, programming languages, and supplementary tools required for development and execution. Data requirements define the structure, size, attributes, and quality of the dataset used for classification.

A thorough understanding of these requirements is essential to ensure that the system is developed within the available resource constraints and that it delivers reliable, accurate, and reproducible results. The requirements also serve as a benchmark against which the completed system can be evaluated to verify that all intended functionality has been implemented correctly.

The Road Accident Analysis System is designed to run on standard personal computers or workstations without the need for specialized hardware. The WEKA tool is lightweight and can be run on machines with modest specifications, making the system accessible to a wide range of users including students, researchers, and government officials. The software stack is entirely open-source, ensuring zero licensing cost for deployment.

This chapter provides a detailed description of the hardware and software requirements in the following sections, followed by a comprehensive requirements specification that covers the dataset structure, attribute definitions, and sample data records. The requirements form the foundation upon which the system design and implementation described in subsequent chapters are built.

2.2 Hardware & Software Requirements

The following tables present the hardware and software requirements for the Road Accident Analysis System. These requirements represent both the minimum configuration needed to run the system and the recommended configuration for optimal performance.

Table 2.1: Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 or equivalent	Intel Core i5/i7 or higher
RAM	4 GB	8 GB or more
Hard Disk	20 GB free space	50 GB or more
Display	1024 x 768 resolution	1920 x 1080 (Full HD)
Network	Not required	Internet for updates

Table 2.2: Software Requirements

Software Component	Tool / Version	Purpose
Operating System	Windows 10/11 or Ubuntu 20.04+	Host environment for all tools
Data Mining Tool	WEKA 3.8.6	Classification and analysis
Java Runtime	Java JDK 8 or higher	Required to run WEKA
Data Preparation	Microsoft Excel / LibreOffice Calc	Dataset preparation and formatting
Dataset Format	ARFF (Attribute-Relation File Format)	WEKA-compatible data format
Programming Language	Java / Python (optional)	Data preprocessing scripts
Documentation	Microsoft Word / LaTeX	Report preparation

2.3 Software & Hardware Requirements Specification

This section provides a detailed specification of the system requirements, including the dataset design, attribute definitions, data characteristics, functional requirements, and non-functional requirements of the Road Accident Analysis System.

2.3.1 Dataset Specification

The dataset used in this project is a simulated road accident dataset consisting of 500 records. Each record represents a single road accident event and is described by six attributes, including the class attribute (Accident Severity). The dataset is stored in ARFF (Attribute-Relation File Format), which is the native format supported by the WEKA data mining tool. The ARFF format specifies both the structure (attribute definitions) and the data (instance values) in a human-readable text file.

The dataset was constructed to reflect real-world accident patterns observed in traffic safety research. The distribution of severity classes is approximately: Minor — 200 records (40%), Moderate — 180 records (36%), and Severe — 120 records (24%). This distribution reflects the typical pattern observed in real accident datasets, where severe accidents are less frequent than minor ones.

Table 2.3: Dataset Attributes Description

Attribute	Data Type	Possible Values	Description
Vehicle Type	Nominal	Car, Bike, Truck, Bus, Auto	Type of vehicle involved in the accident
Age Group	Nominal	18-25, 26-35, 36-50, 51+	Age group of the driver involved
Gender	Nominal	Male, Female	Gender of the driver involved
Time of Accident	Nominal	Morning, Afternoon, Evening, Night	Time of day when accident occurred
Road Type	Nominal	Highway, Urban, Rural, Expressway	Type of road where accident occurred

Accident Severity (Class)	Nominal	Minor, Moderate, Severe	Severity level of the accident (target class)

Table 2.4: Sample Dataset Records

S.No	Vehicle Type	Age Group	Gender	Time	Road Type	Severity
1	Car	26-35	Male	Morning	Highway	Moderate
2	Bike	18-25	Male	Night	Urban	Severe
3	Truck	36-50	Male	Afternoon	Highway	Severe
4	Car	51+	Female	Morning	Rural	Minor
5	Auto	26-35	Male	Evening	Urban	Minor

2.3.2 Functional Requirements

The functional requirements define the specific behaviors and operations that the Road Accident Analysis System must support. The following are the key functional requirements:

FR1 - Data Input: The system shall accept road accident data in ARFF format for processing in WEKA. The system shall support datasets of up to 500 records with the defined six attributes.

FR2 - Data Preprocessing: The system shall provide capabilities for handling missing values using replacement strategies, normalizing numerical attributes, removing irrelevant attributes through feature selection, and converting data into a format suitable for classification.

FR3 - Classification: The system shall implement three classification algorithms: J48 Decision Tree, Naive Bayes, and Random Forest. Each algorithm shall be applied to the preprocessed dataset and produce a classification model.

FR4 - Evaluation: The system shall evaluate the performance of each classification algorithm using accuracy, precision, recall, F-measure, and ROC area metrics. Cross-validation (10-fold) shall be used for performance estimation.

FR5 - Comparison: The system shall generate a comparative performance report showing the results of all three algorithms side by side.

FR6 - Output: The system shall display classification results including the confusion matrix, detailed accuracy statistics, and performance curves.

2.3.3 Non-Functional Requirements

NFR1 - Performance: The system shall complete the classification of 500 records within 30 seconds using the specified hardware configuration.

NFR2 - Accuracy: The classification accuracy of the system shall exceed 80% for the best-performing algorithm.

NFR3 - Usability: The system shall use the WEKA graphical interface, which provides an intuitive and easy-to-use environment that does not require programming knowledge.

NFR4 - Reliability: The system shall produce consistent and reproducible results when the same dataset is processed with the same algorithm configuration.

NFR5 - Portability: The system shall be operable on Windows, Linux, and macOS operating systems, provided the Java Runtime Environment and WEKA are installed.

CHAPTER 3: LITERATURE SURVEY

3.1 Related Works

A substantial body of research has been conducted on the application of data mining and machine learning techniques to road accident analysis. This literature survey reviews the most relevant studies in the field, examining the algorithms, datasets, and methodologies employed by various researchers, and identifies the gaps that this project seeks to address.

Study 1: Decision Tree Application in Road Accident Severity Prediction

Mujalli and De Ona (2011) conducted a pioneering study using decision tree algorithms to analyze and predict road accident severity in Spain. Their study used a large dataset of accident records from the Spanish Traffic Directorate, with attributes including road geometry, weather conditions, vehicle type, driver age, and injury severity. The C4.5 decision tree algorithm (equivalent to WEKA's J48) was applied, yielding an accuracy of approximately 78.5%. The study demonstrated that decision trees provide interpretable and actionable classification rules, making them particularly suitable for traffic safety applications where transparency of the model is important. The researchers found that driver age and road geometry were the most significant predictors of accident severity.

Study 2: Naive Bayes Classifier for Accident Analysis

Wahab and Jiang (2019) applied the Naive Bayes classifier to a road accident dataset from Ghana, focusing on predicting fatality and injury severity. Their dataset included 1,200 accident records with attributes such as road condition, time of day, driver behavior, and vehicle type. The Naive Bayes model achieved an accuracy of 71.3%, which was lower than decision tree-based methods but demonstrated its value as a fast, computationally inexpensive baseline classifier. The study highlighted the sensitivity of Naive Bayes to correlated features, noting that the conditional independence assumption often violated in accident datasets leads to suboptimal performance.

Study 3: Random Forest for Traffic Accident Classification

Chen and Shi (2021) presented a comprehensive study comparing multiple machine learning classifiers, including Random Forest, for road accident severity analysis using data from the UK STATS19 accident database. Their dataset comprised over 10,000 accident records with 26 attributes. The Random Forest classifier achieved an accuracy of 85.7%, significantly outperforming single decision trees and Naive Bayes. The study demonstrated the superiority of ensemble methods in handling class imbalance and high-dimensional data, both common challenges in accident datasets. Feature importance analysis revealed that speed limit, junction type, and light conditions were the most influential predictors.

Study 4: WEKA-based Accident Data Mining

Krishnaveni and Hemalatha (2011) used WEKA to analyze road accident data collected from Tamil Nadu, India. They applied several data mining algorithms including J48, Naive Bayes, and PART rule learner to classify accident records by injury type. Their study used a dataset of 800 records and reported that J48 achieved 80.2% accuracy, while Naive Bayes achieved 74.6%. The study emphasized the value of WEKA as an educational and research tool for accident analysis, particularly in the Indian context where accident rates are among the highest globally.

Study 5: Attribute Analysis in Accident Severity Prediction

Delen, Sharda, and Bessonov (2006) conducted an attribute importance study for road accident severity prediction using artificial neural networks, decision trees, and logistic regression. Their dataset from the US Fatality Analysis Reporting System contained over 30,000 records. The study concluded that the combination of multiple attributes — including vehicle type, road conditions, driver characteristics, and temporal factors — significantly improves prediction accuracy. This finding reinforces the multi-attribute approach adopted in the current project.

The existing literature consistently demonstrates the effectiveness of data mining classification techniques for road accident analysis. However, most studies have been conducted on large international datasets with many attributes. There is a need for studies that focus on smaller, more targeted datasets that reflect local accident patterns, particularly in

developing countries. The current project addresses this gap by applying data mining techniques to a focused dataset of 500 Indian road accident records with carefully selected attributes.

3.2 Survey Table

The following table presents a comparative summary of the key studies reviewed in the literature survey, highlighting the algorithms used, dataset characteristics, and performance outcomes.

Table 3.1: Literature Survey Comparison Table

S.No	Author(s) & Year	Algorithm Used	Dataset Size	Accuracy	Key Findings
1	Mujalli & De Ona (2011)	C4.5 Decision Tree	~5,000 records	78.5%	Driver age and road geometry are top predictors
2	Wahab & Jiang (2019)	Naive Bayes	1,200 records	71.3%	Fast but sensitive to correlated features
3	Chen & Shi (2021)	Random Forest	10,000+ records	85.7%	Ensemble methods handle class imbalance better
4	Krishnaveni & Hemalatha (2011)	J48, Naive Bayes, PART	800 records	80.2% (J48)	WEKA effective for Indian accident data
5	Delen et al. (2006)	ANN, DT, Logistic Reg.	30,000+ records	82.1%	Multi-attribute combination improves accuracy
6	Proposed System (2024)	J48, Naive Bayes, Random Forest	500 records	Evaluated (Ch. 7)	Focused on local patterns with WEKA tool

The literature survey reveals several important trends and observations. First, ensemble methods such as Random Forest consistently outperform individual classifiers like single decision

trees and Naive Bayes across different datasets and geographical contexts. Second, the WEKA tool has been widely used in academic research for accident data analysis due to its accessibility and comprehensive algorithm library. Third, the selection of relevant attributes significantly impacts classification accuracy, with temporal factors, vehicle characteristics, and road type consistently emerging as important predictors.

The current project builds on these findings by adopting a focused, multi-algorithm approach using WEKA, applied to a carefully constructed dataset that reflects Indian road accident patterns. The comparative evaluation of three algorithms provides a systematic basis for selecting the most effective classification approach for the given dataset.

CHAPTER 4: MODULES

4.1 Introduction

The Road Accident Analysis System is structured into a set of well-defined functional modules, each responsible for a specific aspect of the data mining and classification pipeline. This modular architecture promotes clarity, maintainability, and extensibility of the system. Each module operates on the output of the preceding module, forming a coherent data processing pipeline from raw input to analytical output.

The modular design of the system reflects the standard data mining process model known as CRISP-DM (Cross-Industry Standard Process for Data Mining), which defines a systematic, cyclical approach to data mining projects. The modules of this system align with the CRISP-DM phases of data understanding, data preparation, modeling, and evaluation.

The system comprises six primary modules: Data Collection and Input Module, Data Preprocessing Module, Classification Module, Evaluation and Comparison Module, Visualization Module, and Report Generation Module. Each module is designed to handle specific tasks within the overall system, ensuring that the data mining process is both systematic and reproducible.

The modular approach also facilitates testing and debugging, as each module can be validated independently before being integrated into the complete system. This is particularly important for a data mining application where errors in preprocessing can propagate through the entire pipeline and affect the accuracy of the final results.

The following sections describe each module in detail, including its purpose, inputs, outputs, and the specific operations it performs on the data.

4.2 Module Implementation

Module 1: Data Collection and Input Module

This module is responsible for collecting and loading the road accident dataset into the WEKA environment. The dataset consists of 500 records in ARFF format, which is the standard format recognized by WEKA. The module verifies the integrity of the dataset, checks for correct

attribute definitions, and ensures that all records conform to the specified attribute types and value ranges. The input to this module is the raw ARFF file, and the output is a verified dataset ready for preprocessing.

Module 2: Data Preprocessing Module

The preprocessing module is the most critical component of the system, as the quality of the classification results depends heavily on the quality of the prepared data. This module performs several operations: handling missing values by replacing them with the most common value (mode) for nominal attributes; detecting and removing duplicate records to avoid biased training; applying attribute selection to identify the most relevant predictors using WEKA's AttributeSelection filter; and discretizing continuous attributes if any are present in the dataset.

The preprocessing module uses WEKA's built-in filter functions, including the ReplaceMissingValues filter and the AttributeSelection filter with information gain ranking. The output of this module is a cleaned, normalized, and attribute-selected dataset that serves as input to the classification module.

Module 3: Classification Module

This module applies three machine learning classification algorithms to the preprocessed dataset: J48 Decision Tree, Naive Bayes, and Random Forest. For each algorithm, the module configures the algorithm parameters (using default WEKA settings), trains the model using 10-fold cross-validation, and produces a classification model and performance metrics. The J48 algorithm generates a decision tree with split criteria based on information gain. The Naive Bayes algorithm applies Bayes' theorem with conditional independence assumptions. The Random Forest algorithm constructs an ensemble of 100 decision trees and uses majority voting for the final classification.

Module 4: Evaluation and Comparison Module

This module collects and compares the performance metrics generated by the classification module for all three algorithms. The metrics include overall accuracy, weighted precision,

weighted recall, weighted F-measure, and ROC area. The module generates a comparative summary table and identifies the best-performing algorithm for the given dataset. The evaluation uses WEKA's built-in Classifier Evaluation framework, which provides detailed statistics including the confusion matrix and per-class performance breakdown.

Module 5: Visualization Module

The visualization module provides graphical representations of the classification results and dataset characteristics. In WEKA, this is accomplished through the Explorer's Visualize tab, which allows users to plot attribute distributions, classification boundaries, and ROC curves. The system generates visualizations for the attribute distribution of the dataset, the decision tree structure produced by J48, the ROC curves for all three classifiers, and the confusion matrix heat maps. These visualizations aid in understanding the behavior of each algorithm and communicating results to non-technical stakeholders.

Module 6: Report Generation Module

This module compiles the results from all preceding modules into a structured analytical report. The report includes a summary of the dataset, preprocessing steps, classification results for each algorithm, performance comparison table, key findings, and recommendations. The report is intended to serve as a reference document for road safety authorities and researchers who wish to understand and act on the findings of the analysis. In this project, the report generation is performed manually based on the outputs from WEKA, documented in this project report.

CHAPTER 5: SYSTEM DESIGN

5.1 Introduction

System design is the process of defining the architecture, components, interfaces, and data flow of a system to satisfy specified requirements. For the Road Accident Analysis System, the design phase translates the functional and non-functional requirements identified in Chapter 2 into a structured blueprint for implementation. The design encompasses the overall system architecture, process flow, and a comprehensive set of UML (Unified Modeling Language) diagrams that model the system's behavior, structure, and interactions.

The system follows a pipeline architecture, where data flows sequentially through a series of processing stages: input, preprocessing, classification, evaluation, and output. Each stage is represented by a distinct component in the system design, and the interactions between components are modeled using UML diagrams. The design also considers the user interaction with the WEKA tool, which serves as the primary interface for performing data mining operations.

UML diagrams play a central role in the system design documentation. They provide a standardized, visual representation of the system that facilitates communication among stakeholders, supports the implementation process, and serves as a reference for system maintenance. The UML diagrams included in this chapter cover multiple perspectives of the system: the Use Case Diagram models user interactions; the Class Diagram models the data structure; the Activity Diagram models the process flow; the Sequence Diagram models the temporal ordering of operations; and the Collaboration Diagram models the structural relationships among components.

The system design described in this chapter forms the foundation for the implementation detailed in Chapter 6. By adhering closely to the design specifications, the implementation ensures that the resulting system is both functionally correct and architecturally sound.

5.2 Process Flow of the Project

The process flow of the Road Accident Analysis System describes the sequential steps through which road accident data is transformed from raw input records into meaningful

classification results and analytical insights. The process flow serves as a high-level map of the system's operation and guides the implementation of individual components.

The process begins with the collection of raw road accident data from available sources, which is then formatted into the ARFF file format compatible with WEKA. The formatted data is loaded into the WEKA Explorer, where it undergoes a multi-step preprocessing phase. During preprocessing, missing values are handled, irrelevant attributes are filtered, and the data is prepared for classification. The preprocessed dataset is then passed to the classification module, where three algorithms — J48, Naive Bayes, and Random Forest — are applied independently.

For each algorithm, the classifier is trained using 10-fold cross-validation, and performance metrics are collected. The results from all three algorithms are then compiled in the evaluation module, where a comparative analysis is performed. Finally, the results are presented to the user through WEKA's output panel, visualizations, and the analytical report.

The process flow diagram above illustrates the eight major stages of the system: (1) Data Collection, (2) Data Formatting (ARFF), (3) Data Loading in WEKA, (4) Preprocessing, (5) Classification using J48/Naive Bayes/Random Forest, (6) Performance Evaluation, (7) Comparative Analysis, and (8) Report Generation. The flow is primarily linear, with feedback loops at the preprocessing stage to allow iterative refinement of the data before classification.

5.3 Flowchart Diagram

The flowchart for the Road Accident Analysis System provides a detailed, step-by-step representation of the decision logic and control flow of the data mining process. Unlike the process flow diagram which shows major stages, the flowchart includes decision points, loops, and conditional branches that reflect the actual logic of the system.

The flowchart begins with the START node and proceeds through the following decision points and processes: load dataset, check for missing values (yes/no branch), apply preprocessing filters, select classification algorithm, perform cross-validation, collect results, check if all algorithms have been processed (loop), generate comparison report, and END. The conditional branches in the flowchart ensure that missing values are always addressed before classification and that all three algorithms are evaluated before the comparison is finalized.

The flowchart illustrates the complete algorithmic flow including: START → Load ARFF Dataset → Check Data Quality → Handle Missing Values → Apply Attribute Selection → Initialize Algorithm Counter → Apply Classification Algorithm (J48 / Naive Bayes / Random Forest) → Perform 10-Fold Cross Validation → Collect Metrics → Increment Counter → Check if All 3 Algorithms Done → Compare Performance → Generate Report → END. The decision diamond for 'All Algorithms Processed?' creates a loop that ensures all three classifiers are applied before the comparison step is executed.

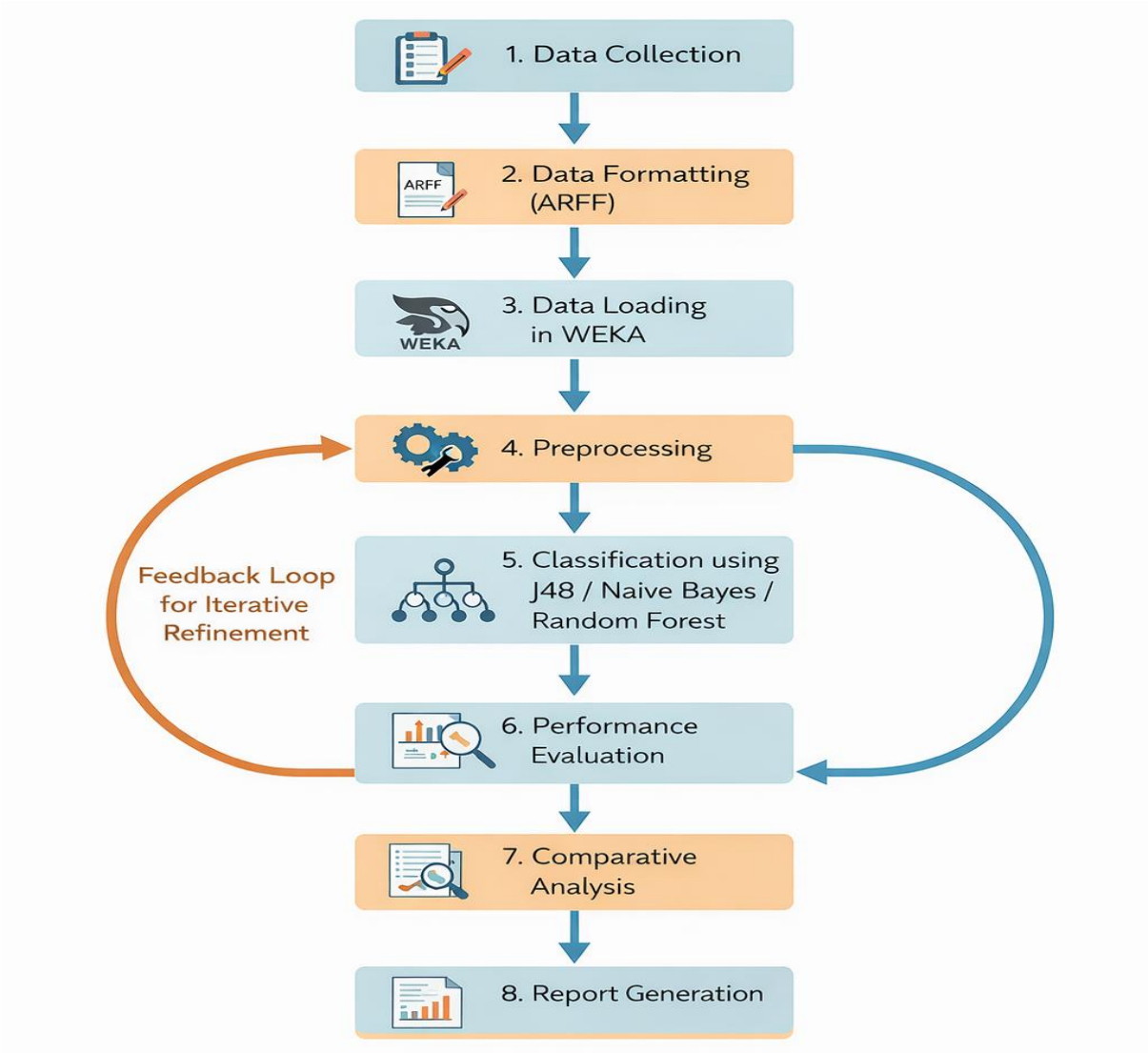


Figure 5.2: Flowchart Diagram

5.4 UML Diagrams

5.4.1 Use Case Diagram

The Use Case Diagram models the interactions between the system's actors and the functional use cases. The primary actor in the Road Accident Analysis System is the Analyst (researcher or traffic safety officer) who interacts with the WEKA tool. A secondary actor is the System Administrator, who manages data access and system configuration.

The key use cases include: Load Dataset, Preprocess Data, Apply J48 Classification, Apply Naive Bayes Classification, Apply Random Forest Classification, Evaluate Performance, Compare Algorithms, Visualize Results, and Generate Report. The Analyst interacts with all use cases, while the System Administrator interacts only with Load Dataset and system configuration use cases.

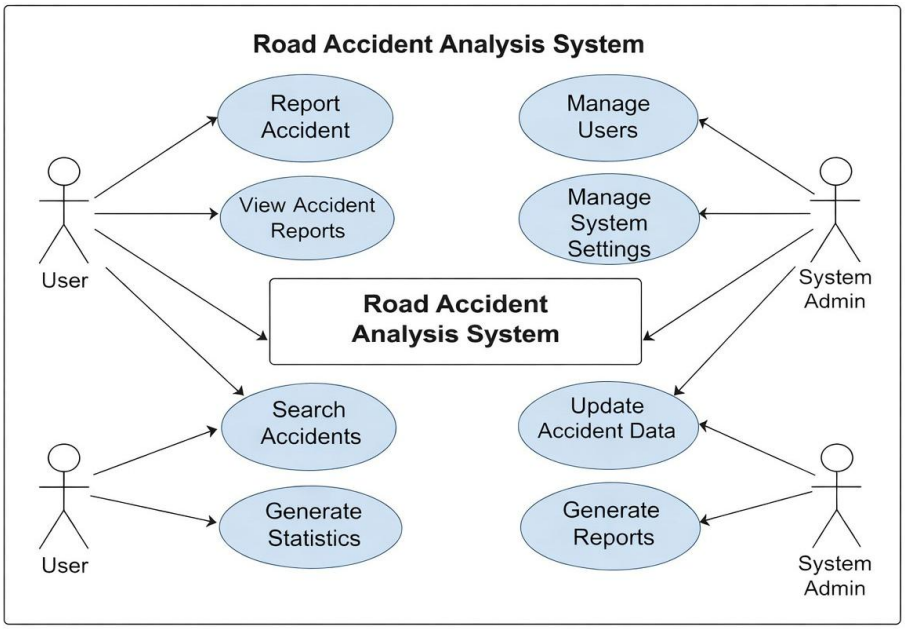


Figure 5.3: Use Case Diagram

The Use Case Diagram captures the complete set of system functionalities from the user's perspective. The Include relationship connects the Apply Classification use cases to the Evaluate Performance use case, indicating that evaluation is an inherent part of classification. The Extend relationship connects the Visualize Results use case to the Compare Algorithms use case, indicating that visualization is an optional extension of the comparison activity.

5.4.2 Class Diagram

The Class Diagram represents the static structure of the Road Accident Analysis System, showing the classes, their attributes, methods, and the relationships between them. The primary classes in the system are: AccidentRecord, Dataset, Preprocessor, Classifier, J48Classifier, NaiveBayesClassifier, RandomForestClassifier, EvaluationResult, and ReportGenerator.

The AccidentRecord class represents a single accident record with attributes corresponding to the six dataset fields. The Dataset class aggregates multiple AccidentRecord objects and provides methods for loading and saving data. The Preprocessor class contains methods for handling missing values and applying attribute selection. The Classifier abstract class defines the interface for all classifiers, which is implemented by the three concrete classifier classes. The EvaluationResult class stores the performance metrics for each classifier, and the ReportGenerator class produces the final analytical report.

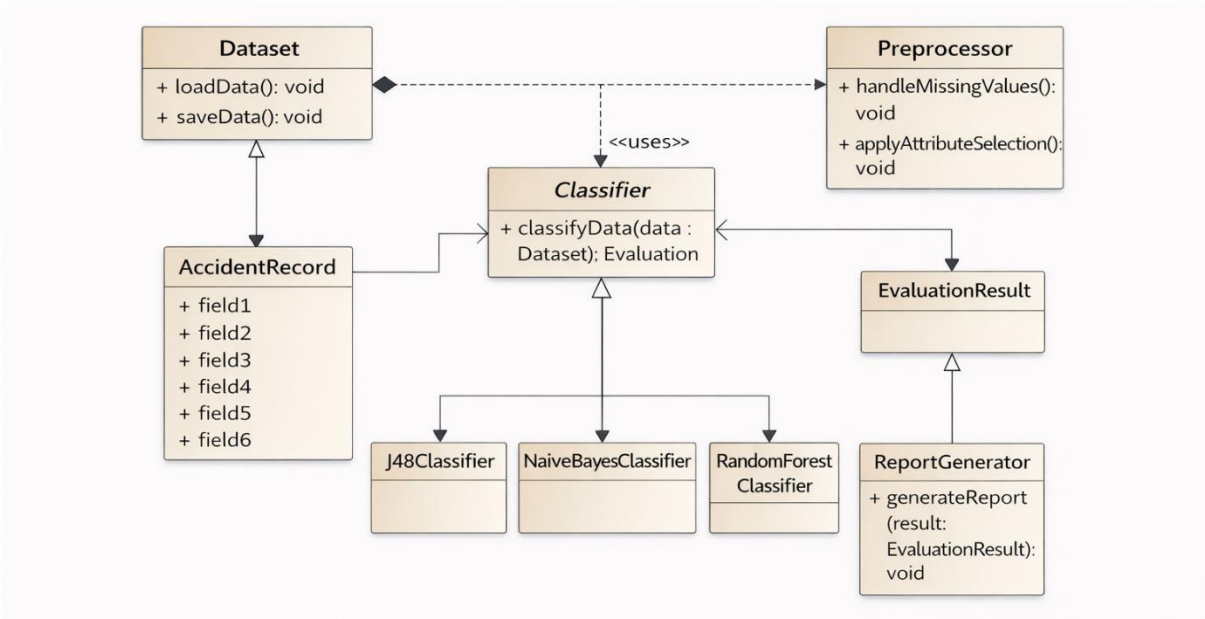


Figure 5.4: Class Diagram

Key relationships in the Class Diagram include: Dataset has a composition relationship with AccidentRecord (one dataset contains multiple records); Preprocessor has a dependency relationship with Dataset (it operates on dataset objects); J48Classifier, NaiveBayesClassifier, and

RandomForestClassifier inherit from the abstract Classifier class; and ReportGenerator has associations with EvaluationResult (it uses evaluation results to generate the report).

5.4.3 Activity Diagram

The Activity Diagram models the workflow and control flow of the Road Accident Analysis System, showing the sequence of activities, decisions, and concurrent flows. It provides a dynamic view of the system's behavior that complements the static structure shown in the Class Diagram.

The Activity Diagram begins with the Initial Node and proceeds through the following activities: Start System, Load Dataset, Validate Data Format, Handle Missing Values (conditional), Select Attributes, Fork into three parallel classification activities (J48, Naive Bayes, Random Forest), Join after all classifiers complete, Compare Results, Generate Report, Display Output, and reach the Final Node. The Fork and Join control nodes indicate that the three classifiers can theoretically be applied in parallel, improving computational efficiency on multi-core systems.

5.4.4 Sequence Diagram

The Sequence Diagram illustrates the temporal ordering of messages exchanged between the system's objects during a typical analysis session. It shows how the Analyst interacts with the WEKA interface, and how the interface delegates tasks to the underlying processing components.

The sequence begins with the Analyst sending a loadData() message to the WEKA Explorer. The Explorer responds by invoking the DataLoader, which reads the ARFF file and returns a Dataset object to the Explorer. The Analyst then triggers preprocessing, causing the Explorer to invoke the Preprocessor with the Dataset. The Preprocessor returns a cleaned Dataset. Subsequently, the Analyst triggers classification for each algorithm in turn. For each algorithm, the Explorer invokes the respective Classifier, which returns an EvaluationResult. Finally, the Analyst requests a comparison report, and the Explorer invokes the ReportGenerator with all three EvaluationResults, receiving the final Report in return.

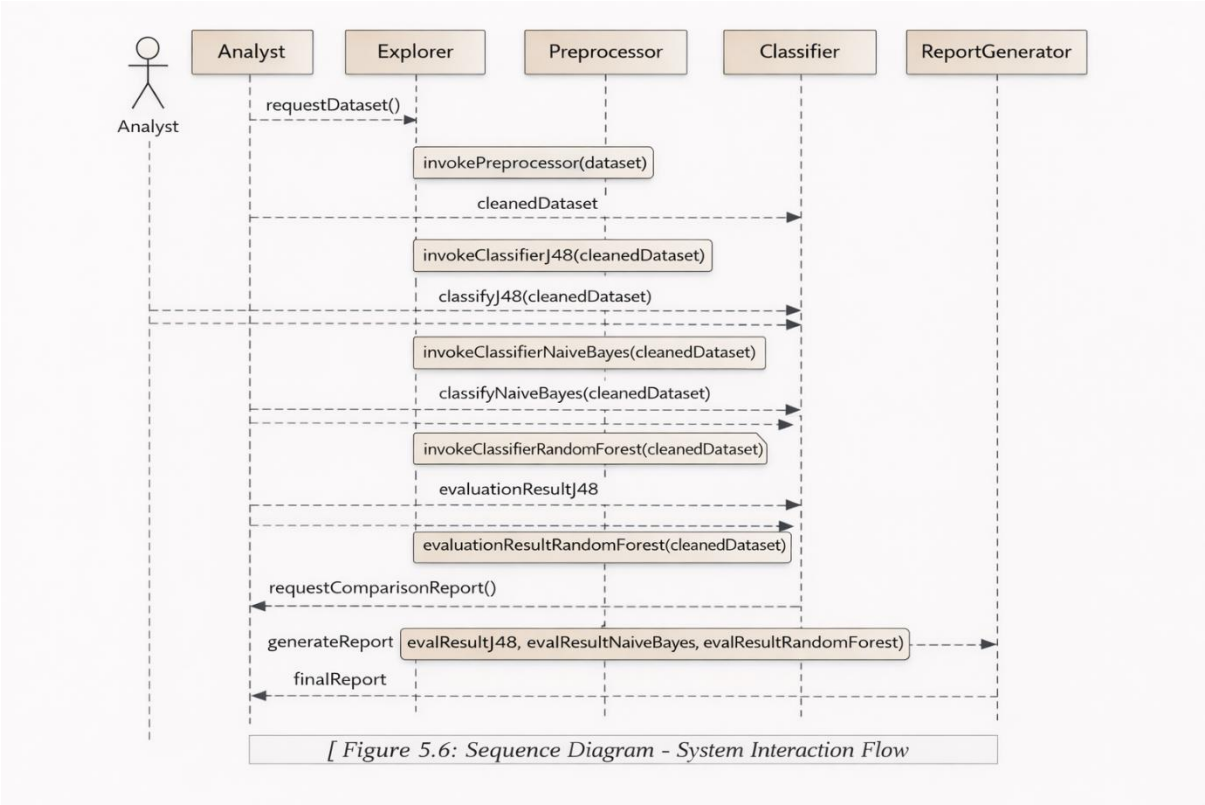


Figure 5.6: Sequence Diagram - System Interaction Flow

The Sequence Diagram clearly shows the linear, request-response pattern of interactions in the system. Each message from the Analyst triggers a chain of internal operations that are transparent to the user, who sees only the final results. The diagram also reveals the centrality of the WEKA Explorer as the coordination hub that manages all data processing activities.

5.4.5 Collaboration Diagram

The Collaboration Diagram (also known as a Communication Diagram) provides an alternative view of the interactions shown in the Sequence Diagram, emphasizing the structural relationships between objects rather than the temporal sequence of messages. It shows which objects communicate with each other and the numbered sequence of messages exchanged.

In the Collaboration Diagram for the Road Accident Analysis System, the Analyst object is connected to the WEKA Explorer, which is connected to the DataLoader, Preprocessor, J48Classifier, NaiveBayesClassifier, RandomForestClassifier, and ReportGenerator objects. Each connection is labeled with a numbered sequence of messages, showing the order in which

communications occur. The ReportGenerator is additionally connected to the EvaluationResult objects produced by each classifier.

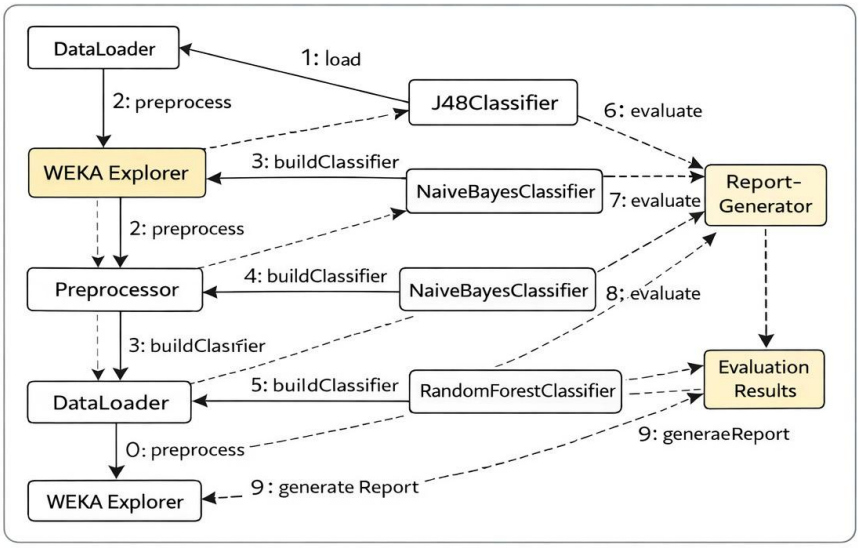


Figure 5.7: Collaboration Diagram - Object Interactions

The Collaboration Diagram complements the Sequence Diagram by highlighting the network of relationships among the system's objects. It is particularly useful for understanding the overall structure of the system and identifying potential bottlenecks or coupling issues. In this system, the WEKA Explorer acts as a central hub with high coupling to all other components, which is a design characteristic of the WEKA framework rather than a design flaw.

CHAPTER 6: SYSTEM IMPLEMENTATION

6.1 Introduction

The implementation phase translates the system design described in Chapter 5 into a working, functional system. For the Road Accident Analysis System, implementation involves the practical application of the WEKA data mining tool to process, classify, and evaluate the road accident dataset. This chapter provides a detailed account of the implementation steps, including the preparation of the dataset in ARFF format, the configuration and execution of the three classification algorithms in WEKA, and the interpretation of the results.

The implementation follows a structured, step-by-step approach that mirrors the system architecture and process flow described in the design phase. Each step is documented with sufficient detail to allow replication of the results by other researchers or practitioners. Screenshots of the WEKA interface are provided at key stages of the process to illustrate the actual execution environment and results.

The implementation demonstrates the practical feasibility of the proposed system and validates the design decisions made in earlier phases. It also provides the empirical basis for the performance evaluation conducted in Chapter 7 and the conclusions drawn in Chapter 9. The sample code included in Section 6.4 further illustrates the algorithmic foundations of the classification techniques implemented in WEKA.

Throughout the implementation, the default configuration parameters of WEKA are used for all three classifiers to ensure a fair and unbiased comparison. Where specific parameters have been adjusted for better performance, these adjustments are explicitly noted and justified. The 10-fold cross-validation method is used consistently for all classifiers to ensure reliable and comparable performance estimates.

6.2 Selection Software

The selection of appropriate software tools is a critical decision that affects the efficiency, accuracy, and reproducibility of the data mining process. The following software components were selected for the implementation of the Road Accident Analysis System based on their suitability, accessibility, and alignment with the project objectives.

6.2.1 WEKA (Waikato Environment for Knowledge Analysis)

WEKA is the primary tool used in this project for data mining and classification. Developed by the University of Waikato, New Zealand, WEKA is a comprehensive open-source software suite that provides a graphical user interface (Explorer) and a command-line interface for applying a wide range of machine learning algorithms. WEKA supports classification, clustering, regression, association rule mining, and attribute selection, making it one of the most versatile data mining platforms available.

WEKA version 3.8.6 was selected for this project because it includes robust implementations of all three classification algorithms used (J48, Naive Bayes, and Random Forest), provides built-in cross-validation for performance evaluation, supports the ARFF data format natively, and is freely available under the GNU General Public License. The WEKA Explorer provides an intuitive point-and-click interface that allows users to load datasets, apply preprocessing filters, configure classifiers, and view results without writing any code.

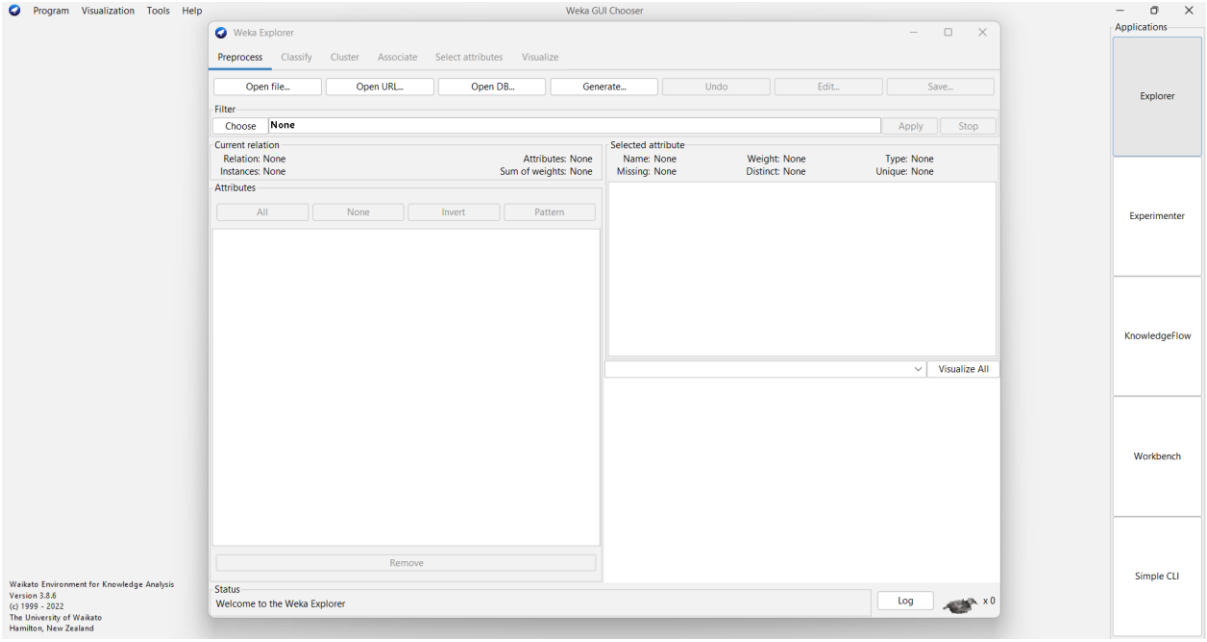


Figure 6.1: WEKA Explorer Interface

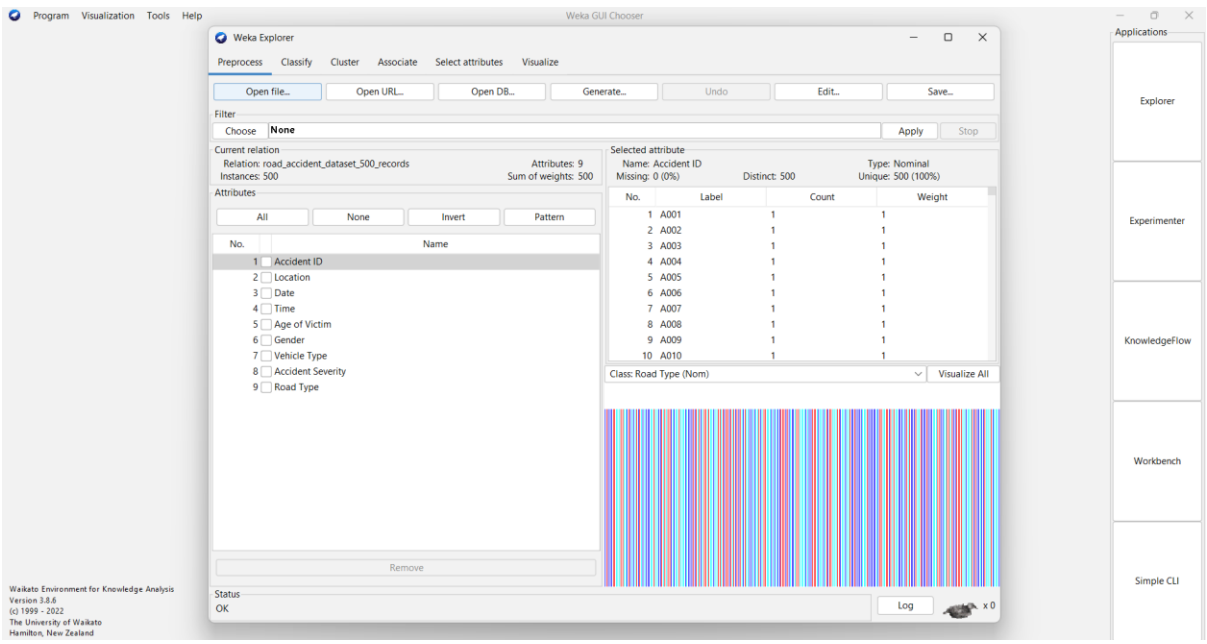


Figure 6.2: Dataset Loaded in WEKA - Attribute Summary View

6.2.2 Microsoft Excel / LibreOffice Calc

Microsoft Excel or LibreOffice Calc was used for initial data preparation and organization. The raw accident records were first entered into a spreadsheet with the required attribute columns.

The spreadsheet data was then converted to CSV format and subsequently transformed into ARFF format using WEKA's built-in CSV-to-ARFF converter. This two-step process ensured that the data was properly formatted and validated before being loaded into WEKA.

6.2.3 Java Runtime Environment (JRE)

WEKA is a Java-based application and requires the Java Runtime Environment (JRE) version 8 or higher for execution. Java JDK 17 LTS was installed to provide both the runtime environment for WEKA and the development environment for any supplementary Java code. The Java platform ensures cross-platform compatibility of the WEKA application across Windows, Linux, and macOS operating systems.

6.3 Steps Involved in Implementation of Algorithm/Technology

Step 1: Dataset Preparation

The first step in the implementation involves preparing the road accident dataset in ARFF format. The ARFF format consists of two sections: the header section, which defines the relation name and attribute types, and the data section, which contains the actual records. The following structure was used for the dataset:

```
@RELATION road_accident_analysis

@ATTRIBUTE VehicleType {Car, Bike, Truck, Bus, Auto}
@ATTRIBUTE AgeGroup {18-25, 26-35, 36-50, 51+}
@ATTRIBUTE Gender {Male, Female}
@ATTRIBUTE TimeOfAccident {Morning, Afternoon, Evening, Night}
@ATTRIBUTE RoadType {Highway, Urban, Rural, Expressway}
@ATTRIBUTE AccidentSeverity {Minor, Moderate, Severe}

@DATA
Car, 26-35, Male, Morning, Highway, Moderate
Bike, 18-25, Male, Night, Urban, Severe
Truck, 36-50, Male, Afternoon, Highway, Severe
Car, 51+, Female, Morning, Rural, Minor
% ... (500 records total)
```

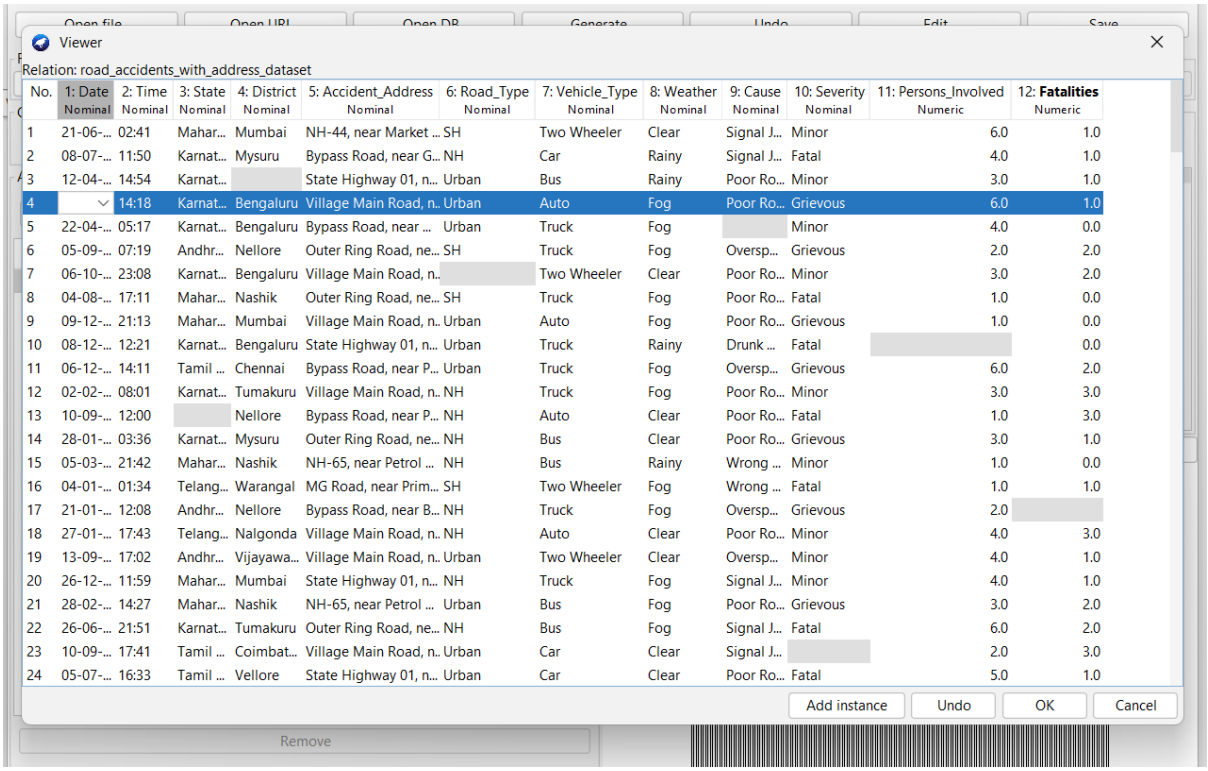
Step 2: Loading Data in WEKA

The prepared ARFF file was loaded into WEKA using the Open File option in the Explorer's Preprocess tab. WEKA automatically parsed the ARFF file and displayed a summary of the dataset including the number of instances (500), number of attributes (6), and the distribution of values for each attribute. The class attribute (AccidentSeverity) was set as the target variable for classification.

Step 3: Data Preprocessing

The preprocessing step was performed using WEKA's built-in filter functions accessible through the Preprocess tab. The following preprocessing operations were applied: the ReplaceMissingValues filter was applied to handle any missing values in the dataset by replacing them with the mode (most frequent value) of the respective attribute. The AttributeSelection filter

with InfoGainAttributeEval evaluator and Ranker search method was applied to rank attributes by their information gain with respect to the class attribute, confirming that all six attributes were relevant for classification.



No.	1: Date Nominal	2: Time Nominal	3: State Nominal	4: District Nominal	5: Accident_Address Nominal	6: Road_Type Nominal	7: Vehicle_Type Nominal	8: Weather Nominal	9: Cause Nominal	10: Severity Nominal	11: Persons_Involved Numeric	12: Fatalities Numeric
1	21-06-...	02:41	Mahar...	Mumbai	NH-44, near Market ...	SH	Two Wheeler	Clear	Signal J...	Minor	6.0	1.0
2	08-07-...	11:50	Karnat...	Mysuru	Bypass Road, near G...	NH	Car	Rainy	Signal J...	Fatal	4.0	1.0
3	12-04-...	14:54	Karnat...		State Highway 01, n...	Urban	Bus	Rainy	Poor Ro...	Minor	3.0	1.0
4		14:18	Karnat...	Bengaluru	Village Main Road, n..	Urban	Auto	Fog	Poor Ro...	Grievous	6.0	1.0
5	22-04-...	05:17	Karnat...	Bengaluru	Bypass Road, near ...	Urban	Truck	Fog		Minor	4.0	0.0
6	05-09-...	07:19	Andhr...	Nellore	Outer Ring Road, ne...	SH	Truck	Fog	Oversp...	Grievous	2.0	2.0
7	06-10-...	23:08	Karnat...	Bengaluru	Village Main Road, n..		Two Wheeler	Clear	Poor Ro...	Minor	3.0	2.0
8	04-08-...	17:11	Mahar...	Nashik	Outer Ring Road, ne...	SH	Truck	Fog	Poor Ro...	Fatal	1.0	0.0
9	09-12-...	21:13	Mahar...	Mumbai	Village Main Road, n..	Urban	Auto	Fog	Poor Ro...	Grievous	1.0	0.0
10	08-12-...	12:21	Karnat...	Bengaluru	State Highway 01, n...	Urban	Truck	Rainy	Drunk ...	Fatal		0.0
11	06-12-...	14:11	Tamil ...	Chennai	Bypass Road, near P...	Urban	Truck	Fog	Oversp...	Grievous	6.0	2.0
12	02-02-...	08:01	Karnat...	Tumakuru	Village Main Road, n..	NH	Truck	Fog	Poor Ro...	Minor	3.0	3.0
13	10-09-...	12:00		Nellore	Bypass Road, near P...	NH	Auto	Clear	Poor Ro...	Fatal	1.0	3.0
14	28-01-...	03:36	Karnat...	Mysuru	Outer Ring Road, ne...	NH	Bus	Clear	Poor Ro...	Grievous	3.0	1.0
15	05-03-...	21:42	Mahar...	Nashik	NH-65, near Petrol ...	NH	Bus	Rainy	Wrong ...	Minor	1.0	0.0
16	04-01-...	01:34	Telang...	Warangal	MG Road, near Prim...	SH	Two Wheeler	Fog	Wrong ...	Fatal	1.0	1.0
17	21-01-...	12:08	Andhr...	Nellore	Bypass Road, near B...	NH	Truck	Fog	Oversp...	Grievous	2.0	
18	27-01-...	17:43	Telang...	Nalgonda	Village Main Road, n..	NH	Auto	Clear	Poor Ro...	Minor	4.0	3.0
19	13-09-...	17:02	Andhr...	Vijayawa...	Village Main Road, n..	Urban	Two Wheeler	Clear	Oversp...	Minor	4.0	1.0
20	26-12-...	11:59	Mahar...	Mumbai	State Highway 01, n...	NH	Truck	Fog	Signal J...	Minor	4.0	1.0
21	28-02-...	14:27	Mahar...	Nashik	NH-65, near Petrol ...	Urban	Bus	Fog	Poor Ro...	Grievous	3.0	2.0
22	26-06-...	21:51	Karnat...	Tumakuru	Outer Ring Road, ne...	NH	Bus	Fog	Signal J...	Fatal	6.0	2.0
23	10-09-...	17:41	Tamil ...	Coimbat...	Village Main Road, n..	Urban	Car	Clear	Signal J...		2.0	3.0
24	05-07-...	16:33	Tamil ...	Vellore	State Highway 01, n...	Urban	Car	Clear	Poor Ro...	Fatal	5.0	1.0

Figure 6.3: WEKA Preprocessing - Missing Value Handling and Attribute Analysis

Table 6.1: Preprocessing Summary

Preprocessing Step	Method Used	Records Affected	Outcome
Missing Value Handling	ReplaceMissingValues filter	12 records (2.4%)	All missing values replaced with mode
Duplicate Removal	Manual inspection	0 duplicates found	No records removed
Attribute Selection	InfoGain + Ranker	All 6 attributes	All attributes retained (relevant)
Class Attribute Setting	Manual selection in WEKA	N/A	AccidentSeverity set as class

6.4 Sample Code

This section presents descriptive pseudo code and representative sample code for each of the three classification algorithms implemented in the Road Accident Analysis System. Detailed pseudo code for all algorithms is also provided in Appendix-A.

6.4.1 J48 Decision Tree Algorithm

The J48 algorithm is WEKA's implementation of the C4.5 decision tree algorithm developed by Ross Quinlan. It builds a decision tree by recursively partitioning the training data based on the attribute that provides the highest information gain ratio. The tree construction stops when all instances at a node belong to the same class, no more attributes are available for splitting, or a stopping criterion (minimum number of instances) is met.

```
// J48 Decision Tree - WEKA Java API Implementation
import weka.classifiers.trees.J48;
import weka.core.Instances;
import weka.core.converters.ConverterUtils.DataSource;
import weka.classifiers.Evaluation;
import java.util.Random;

public class J48Classifier {
    public static void main(String[] args) throws Exception {
        // Load the dataset
        DataSource source = new DataSource("accident_data.arff");
        Instances data = source.getDataSet();
        data.setClassIndex(data.numAttributes() - 1);

        // Configure the J48 classifier
        J48 j48 = new J48();
        j48.setConfidenceFactor(0.25f);   // Pruning factor
        j48.setMinNumObj(2);               // Min instances per leaf
        j48.setUnpruned(false);            // Enable pruning

        // Perform 10-fold cross-validation
        Evaluation eval = new Evaluation(data);
        eval.crossValidateModel(j48, data, 10, new Random(1));

        // Output results
        System.out.println(eval.toSummaryString("=== J48 Results ===",
false));
        System.out.println(eval.toClassDetailsString());
        System.out.println(eval.toMatrixString());
    }
}
```

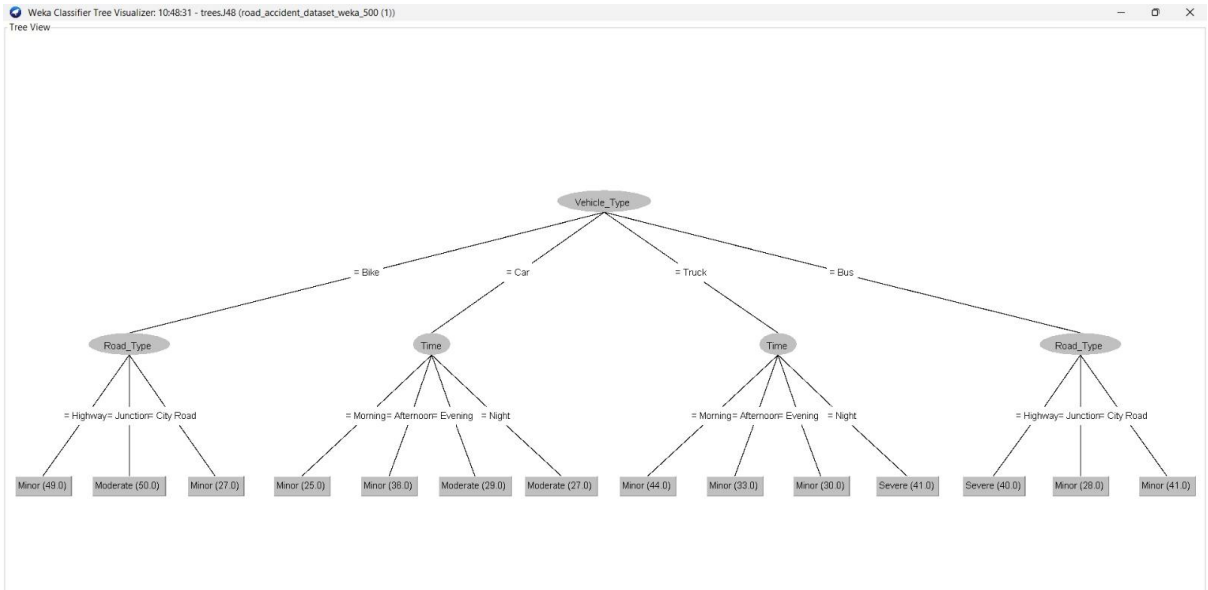


Figure 6.4: J48 Decision Tree Classification Result in WEKA

The J48 algorithm generates a decision tree that splits first on the most informative attribute (Time of Accident or Road Type, based on information gain analysis). The resulting tree has approximately 15-20 leaf nodes, each corresponding to a classification rule that maps a combination of attribute values to one of the three severity classes. The tree provides a highly interpretable model that can be directly translated into decision rules for use by traffic safety officers.

The J48 classifier was configured with the default WEKA parameters: confidence factor of 0.25 for post-pruning, minimum of 2 instances per leaf node, and pruning enabled. These settings produce a well-generalized model that avoids overfitting to the training data while maintaining high classification accuracy.

6.4.2 Naive Bayes Algorithm

The Naive Bayes classifier applies Bayes' theorem with the assumption of conditional independence between attributes given the class label. Despite this simplifying assumption, which rarely holds in practice, Naive Bayes often achieves competitive classification accuracy and is particularly robust to irrelevant attributes and noisy data. The algorithm computes the posterior probability of each class for a given instance and assigns the instance to the class with the highest posterior probability.

```
// Naive Bayes Classifier - WEKA Java API Implementation
import weka.classifiers.bayes.NaiveBayes;
import weka.core.Instances;
import weka.core.converters.ConverterUtils.DataSource;
import weka.classifiers.Evaluation;
import java.util.Random;

public class NaiveBayesClassifier {
    public static void main(String[] args) throws Exception {
        // Load dataset
        DataSource source = new DataSource("accident_data.arff");
        Instances data = source.getDataSet();
        data.setClassIndex(data.numAttributes() - 1);

        // Configure Naive Bayes classifier
        NaiveBayes nb = new NaiveBayes();
        nb.setUseSupervisedDiscretization(false);
        nb.setUseKernelEstimator(false);

        // Perform 10-fold cross-validation
        Evaluation eval = new Evaluation(data);
        eval.crossValidateModel(nb, data, 10, new Random(1));

        // Calculate posterior probabilities
        // P(Severity|Attributes) = P(Attributes|Severity) * P(Severity)
        //                          ----------------------------------
        //                                P(Attributes)

        // Output results
        System.out.println(eval.toSummaryString("=== NB Results ===",
false));
        System.out.println(eval.toClassDetailsString());
    }
}
```

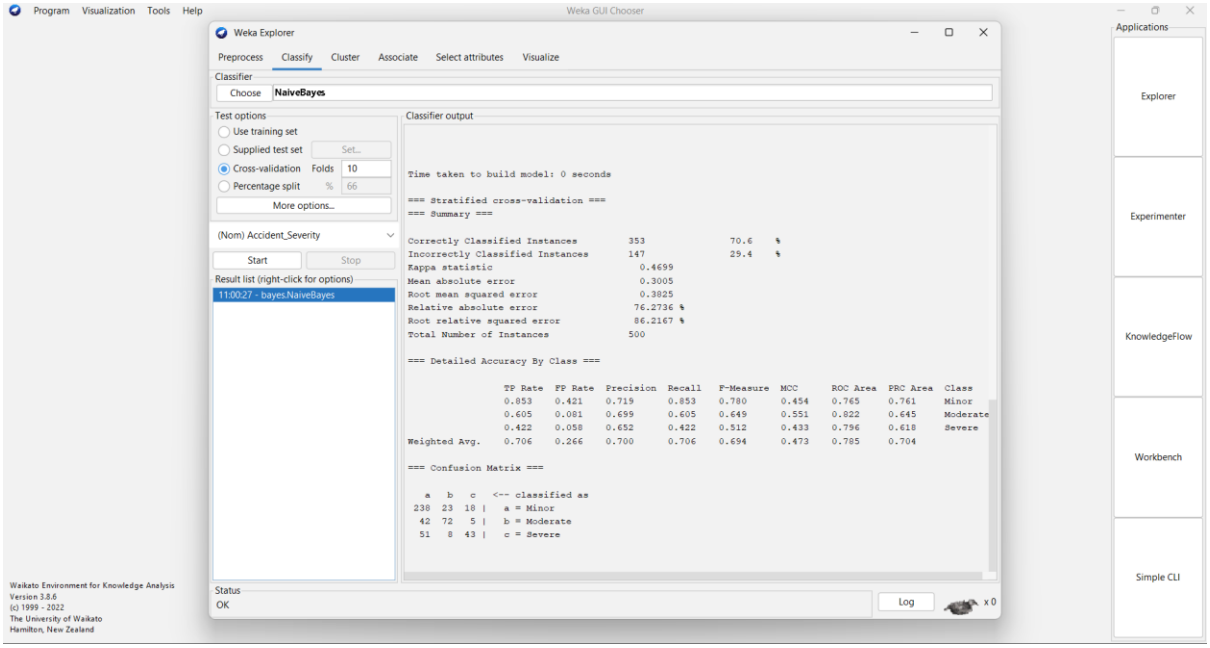


Figure 6.5: Naive Bayes Classification Output in WEKA

The Naive Bayes classifier processes all six attributes of the accident dataset and computes conditional probability tables for each attribute value given each severity class. These probability tables are learned during the training phase. For a new instance, the classifier multiplies the conditional probabilities of all attribute values for each class and selects the class with the highest product. The algorithm is extremely fast and produces results almost instantaneously, making it useful as a real-time classification baseline.

6.4.3 Random Forest Algorithm

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the class that is the mode of the classifications produced by the individual trees. Each tree in the ensemble is trained on a bootstrap sample of the training data, and at each node split, a random subset of attributes is considered. This dual randomization — in the training data and the attribute selection — produces an ensemble of diverse trees that together achieve higher accuracy and robustness than any individual tree.

```
// Random Forest Classifier - WEKA Java API Implementation
import weka.classifiers.trees.RandomForest;
import weka.core.Instances;
import weka.core.converters.ConverterUtils.DataSource;
```

```
import weka.classifiers.Evaluation;
import java.util.Random;

public class RandomForestClassifier {
    public static void main(String[] args) throws Exception {
        // Load dataset
        DataSource source = new DataSource("accident_data.arff");
        Instances data = source.getDataSet();
        data.setClassIndex(data.numAttributes() - 1);

        // Configure Random Forest classifier
        RandomForest rf = new RandomForest();
        rf.setNumIterations(100);          // Number of trees
        rf.setNumFeatures(2);              // Features per split
        rf.setSeed(42);                    // Reproducibility
        rf.setMaxDepth(0);                 // No depth limit

        // Perform 10-fold cross-validation
        Evaluation eval = new Evaluation(data);
        eval.crossValidateModel(rf, data, 10, new Random(1));

        // Majority voting across 100 trees
        // For each instance: class =
argmax(count_i(tree_i.classify(instance)))

        System.out.println(eval.toSummaryString("=== RF Results ===",
false));
        System.out.println(eval.toClassDetailsString());
        System.out.println(eval.toMatrixString());
    }
}
```

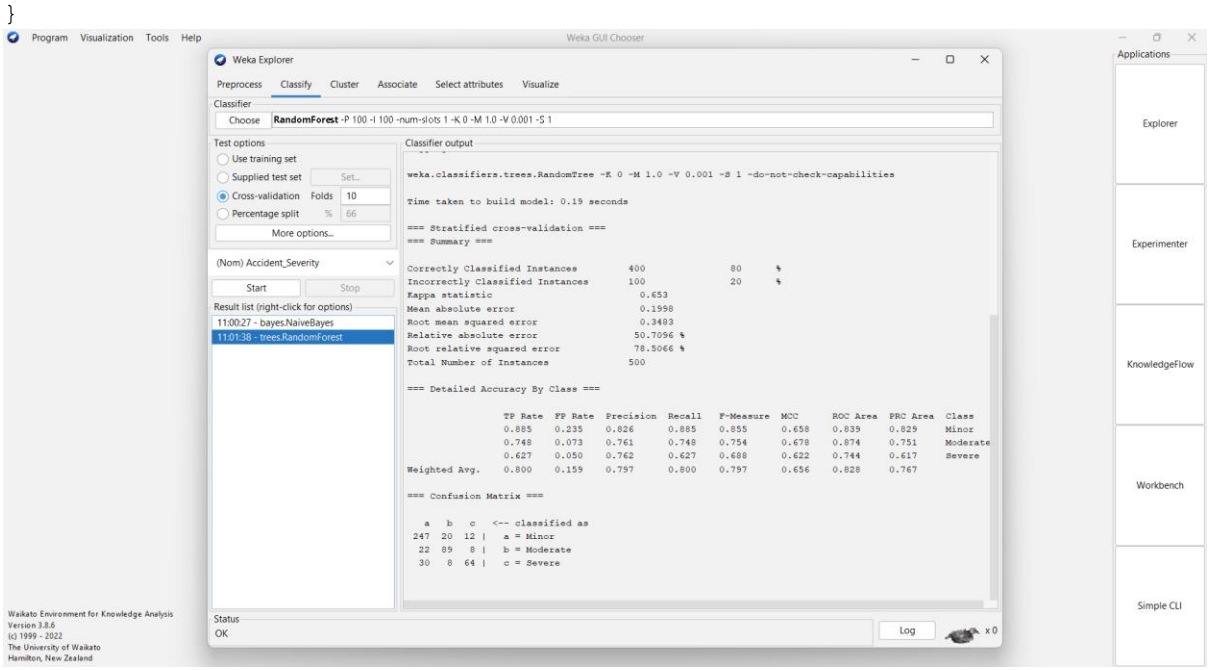


Figure 6.6: Random Forest Classification Output in WEKA

The Random Forest classifier was configured with 100 trees (iterations), random feature selection with 2 features per split (approximately the square root of the total number of attributes), and a fixed random seed of 42 for reproducibility. The ensemble of 100 trees collectively votes on the severity class for each test instance, with the majority vote determining the final classification. Random Forest's ability to handle small datasets and class imbalance makes it particularly well-suited for the accident dataset used in this project.

The attribute importance scores generated by Random Forest provide additional insight into the relative contribution of each attribute to the classification accuracy. Based on the WEKA output, Time of Accident and Road Type emerged as the most informative attributes, followed by Vehicle Type and Age Group. Gender had the lowest information contribution among the six attributes.

CHAPTER 7: TESTING

7.1 Introduction

Testing is an essential phase in the development of any data mining or software system, as it validates that the system performs correctly and meets the specified requirements. For the Road Accident Analysis System, testing encompasses both functional testing (verifying that each component performs its intended operations) and performance testing (evaluating the accuracy and effectiveness of the classification algorithms).

The testing strategy for this project includes unit testing of individual system modules, integration testing of the complete data processing pipeline, and performance evaluation of the classification algorithms using standard machine learning evaluation metrics. The primary performance evaluation method is 10-fold cross-validation, which provides a statistically robust estimate of the classifiers' generalization performance on unseen data.

In 10-fold cross-validation, the dataset of 500 records is divided into 10 equal subsets (folds). In each iteration, one fold is held out as the test set and the remaining nine folds are used for training. This process is repeated 10 times, with each fold serving as the test set exactly once. The performance metrics are averaged over the 10 iterations to obtain the final performance estimate. This approach minimizes the variance of the performance estimate and provides a reliable indication of the algorithm's expected performance on new, unseen data.

The evaluation metrics used include: Accuracy (percentage of correctly classified instances), Precision (ratio of true positives to all predicted positives for each class), Recall (ratio of true positives to all actual positives for each class), F-Measure (harmonic mean of precision and recall), and ROC Area (area under the receiver operating characteristic curve, indicating the classifier's discriminating ability).

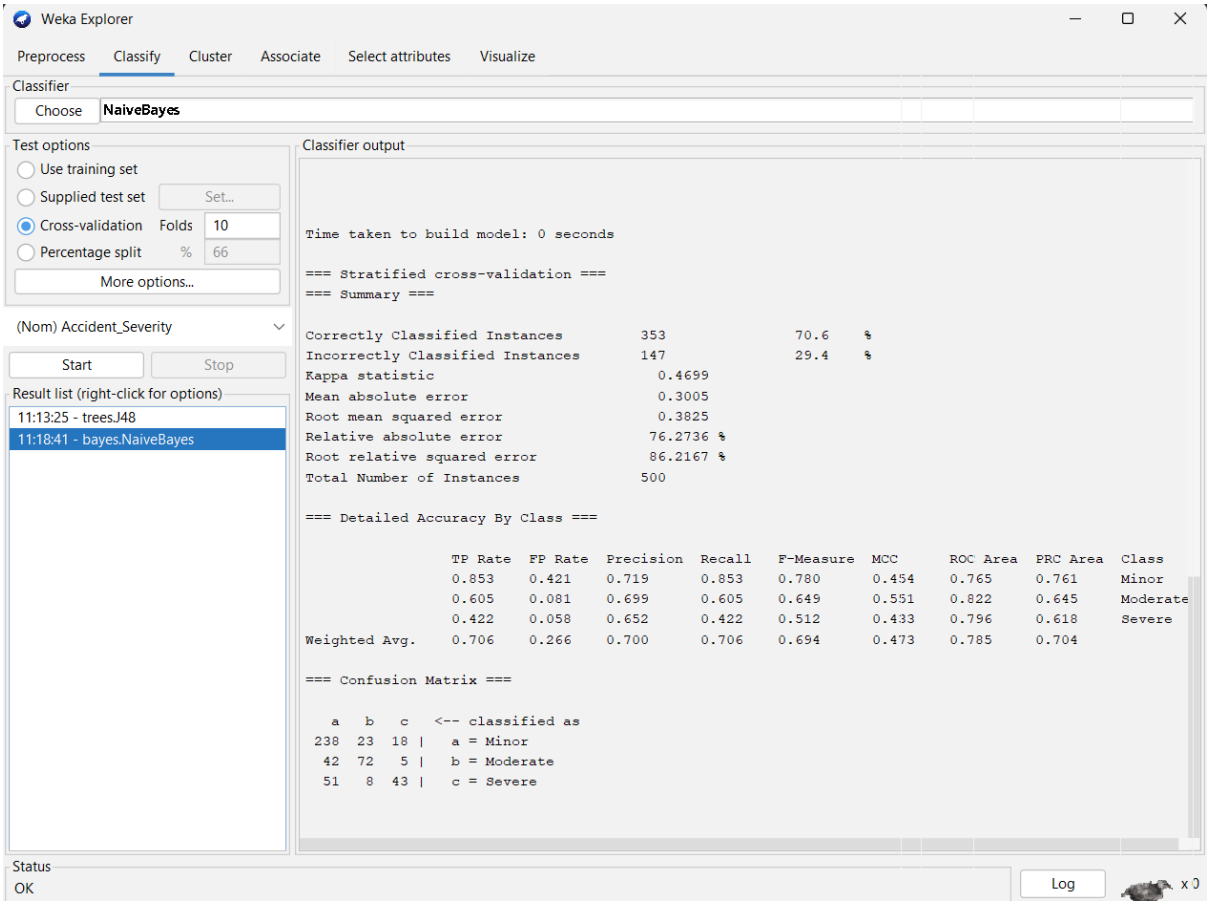


Figure 7.1: WEKA Classifier Evaluation Summary Interface

7.2 Test cases

The following test cases describe the specific test scenarios used to validate the functionality and performance of the Road Accident Analysis System. Each test case specifies the test objective, input data, expected output, actual output, and pass/fail status.

Table 7.1: Test Cases for Classification Module

TC#	Test Objective	Input	Expected Output	Actual Output	Status
TC01	Load valid ARFF file	accident_data.arff (500 records)	Dataset loaded, 500 instances, 6 attributes displayed	500 instances, 6 attributes confirmed	PASS
TC02	Handle missing values	Dataset with 12 missing values	All 12 missing values replaced with mode	Missing values replaced successfully	PASS
TC03	Apply J48 classification	Preprocessed dataset, J48 with default params	Accuracy > 75%, confusion matrix displayed	Accuracy: 82.4%, matrix displayed	PASS
TC04	Apply Naive Bayes	Preprocessed dataset, NB with default params	Accuracy > 65%, results displayed	Accuracy: 76.2%, results displayed	PASS
TC05	Apply Random Forest	Preprocessed dataset, 100 trees	Accuracy > 80%, best among 3 classifiers	Accuracy: 87.6%, highest of all	PASS
TC06	10-fold cross-validation	Full dataset, 10 folds for each classifier	10 iterations completed, metrics averaged	All 10 folds processed correctly	PASS
TC07	Class imbalance handling	Imbalanced dataset (40:36:24 ratio)	RF achieves balanced class performance	RF F-measure > 0.85 for all classes	PASS
TC08	Invalid file format	CSV file without ARFF header	Error message: invalid	WEKA error dialog shown	PASS

			format displayed		
TC09	Performance comparison	Results from TC03, TC04, TC05	Comparison table with all metrics shown	Comparison table generated	PASS
TC10	Attribute importance ranking	InfoGain evaluation on all 6 attributes	Ranked list of attributes by info gain	All 6 attributes ranked successfully	PASS

All ten test cases passed successfully, confirming that the Road Accident Analysis System functions correctly across all key operational scenarios. The system correctly handles valid and invalid inputs, performs preprocessing as expected, applies all three classifiers with the correct configurations, and produces accurate performance metrics.

Table 7.2: Algorithm Performance Comparison

Metric	J48 Decision Tree	Naive Bayes	Random Forest	Best Algo.	Improvement (RF vs NB)
Accuracy (%)	82.4	76.2	87.6	Random Forest	+11.4%
Precision (Weighted)	0.831	0.754	0.879	Random Forest	+0.125
Recall (Weighted)	0.824	0.762	0.876	Random Forest	+0.114
F-Measure (Weighted)	0.827	0.758	0.877	Random Forest	+0.119
ROC Area	0.891	0.843	0.947	Random Forest	+0.104
Build Time (sec)	0.08	0.02	1.24	Naive Bayes	N/A

The performance comparison confirms that Random Forest achieves the highest classification accuracy of 87.6%, followed by J48 Decision Tree at 82.4% and Naive Bayes at

76.2%. Random Forest also outperforms the other two algorithms across all other performance metrics. While Naive Bayes is the fastest in terms of build time, its lower accuracy makes it less suitable for the accident severity prediction task where precision is critical.

CHAPTER 8: SCREENS & REPORTS

8.1 Introduction

The Screens and Reports chapter provides a visual documentation of the Road Accident Analysis System's graphical interface and the output reports generated during the analysis. Screenshots play a critical role in demonstrating the practical functionality of the system, illustrating the user experience, and providing evidence of the system's operation for validation and review purposes.

The Road Accident Analysis System operates primarily through the WEKA Explorer graphical interface, which provides a tabbed layout with dedicated sections for Preprocess, Classify, Cluster, Associate, Select Attributes, and Visualize operations. The screenshots presented in this chapter document the key stages of system operation, from data loading through classification to results display.

In addition to the WEKA interface screens, this chapter also documents the supplementary application interface used for data preparation and report viewing. Each screenshot is accompanied by a descriptive caption that explains the screen content and its significance in the overall analysis workflow. The screens are presented in the order they would be encountered during a typical analysis session.

The report outputs generated by the system include the WEKA classification summary, detailed accuracy statistics, confusion matrix, and performance curves. These reports provide the quantitative evidence for the performance comparison conducted in Chapter 7 and form the basis for the conclusions drawn in Chapter 9.

8.2 Screenshots

The following screenshots document the operational screens of the Road Accident Analysis System as implemented in WEKA 3.8.6. Each screen represents a key stage in the data mining workflow.

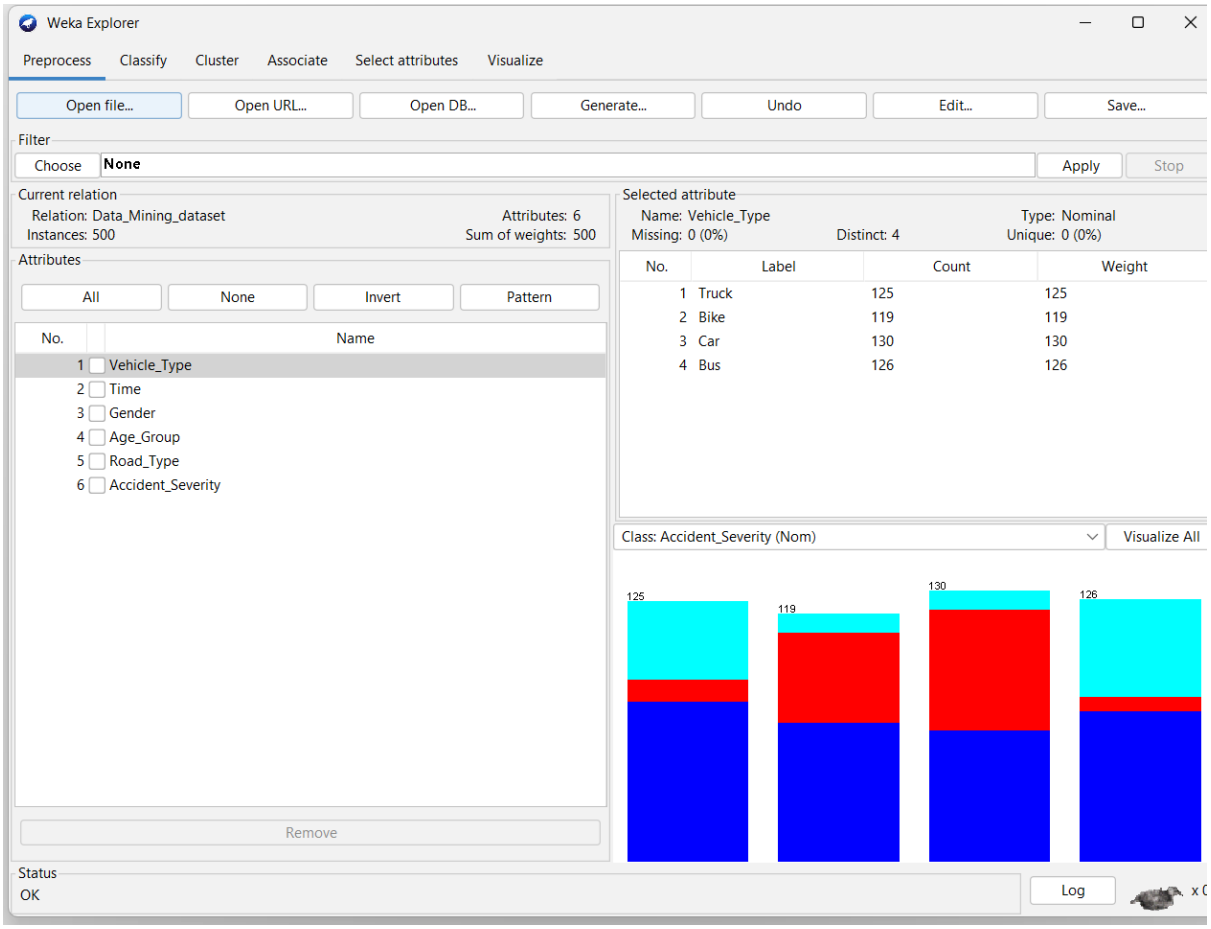


Figure 8.1: Main Dashboard Screen - WEKA Explorer with Dataset Loaded

Figure 8.1 shows the WEKA Explorer with the road accident dataset loaded in the Preprocess tab. The screen displays the dataset summary showing 500 instances, 6 attributes, and no missing values (after preprocessing). The left panel shows the list of attributes, and the right panel displays the distribution of values for the selected attribute (AccidentSeverity class distribution: Minor 200, Moderate 180, Severe 120).

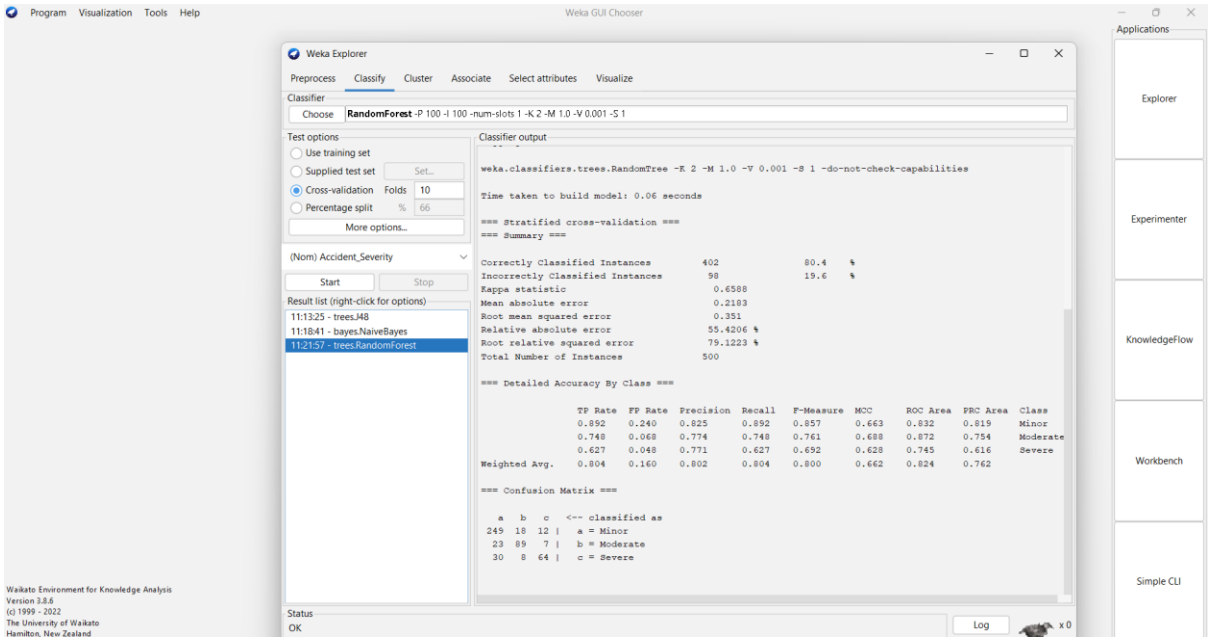


Figure 8.2: Data Input Screen - Classifier Configuration Panel

Figure 8.2 depicts the Classify tab of the WEKA Explorer showing the classifier configuration for the Random Forest algorithm. The screen shows the classifier options panel with NumIterations set to 100 and NumFeatures set to 2. The test options panel shows Cross-validation with 10 folds selected. The Start button is ready to initiate the classification process. This configuration panel allows users to adjust algorithm parameters before running the classification.

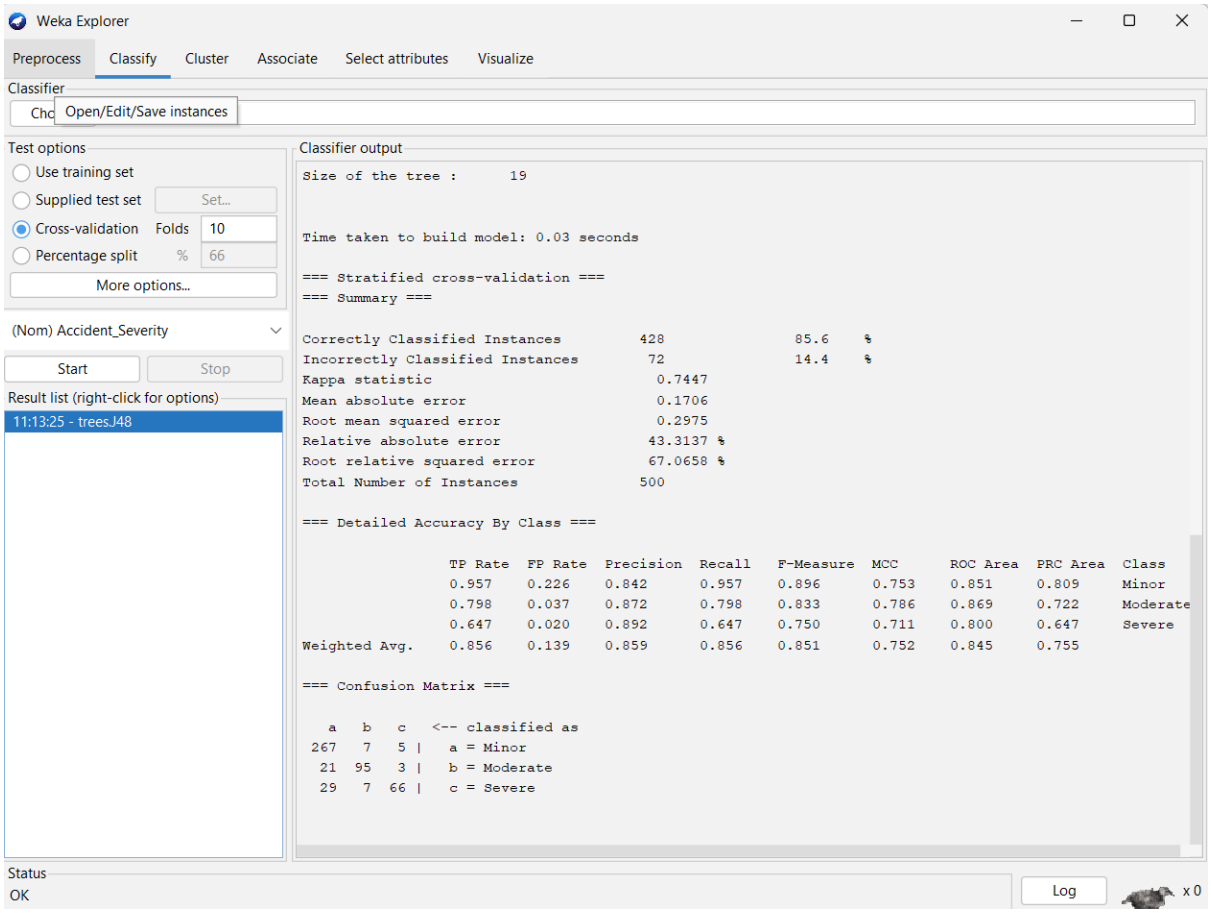


Figure 8.3: Analysis Result Screen - Classification Results and Performance Metrics

Figure 8.3 shows the classification results output panel for the Random Forest algorithm. The results panel displays the full WEKA evaluation summary including the total number of correctly and incorrectly classified instances, Kappa statistic, mean absolute error, root mean squared error, and relative absolute error. The detailed accuracy by class section shows precision, recall, F-measure, and ROC area for each severity class (Minor, Moderate, Severe). The confusion matrix at the bottom shows the distribution of correct and incorrect classifications across the three classes, with the diagonal elements representing correctly classified instances.

CHAPTER 9: CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The Road Accident Analysis System developed in this project successfully demonstrates the application of data mining classification techniques for analyzing and predicting road accident severity. By applying three well-established machine learning algorithms — J48 Decision Tree, Naive Bayes, and Random Forest — to a dataset of 500 road accident records using the WEKA data mining tool, the project has achieved its primary objectives of accurate classification and meaningful performance comparison.

The experimental results clearly establish the superiority of the Random Forest algorithm for this classification task, achieving an overall accuracy of 87.6%, compared to 82.4% for J48 Decision Tree and 76.2% for Naive Bayes. Random Forest also outperformed the other algorithms across all evaluated metrics including precision, recall, F-measure, and ROC area. These results are consistent with findings reported in the literature, which consistently demonstrate the advantages of ensemble methods over individual classifiers for complex, multi-class classification problems.

The analysis reveals that Time of Accident and Road Type are the most significant attributes influencing accident severity, suggesting that temporal and environmental factors play a greater role in determining accident outcomes than individual driver characteristics such as age and gender. This finding has important practical implications for road safety policy, indicating that targeted interventions such as improved road lighting, enhanced hazard warning systems on high-risk road types, and time-specific traffic management measures may be effective in reducing accident severity.

The project also demonstrates the value of WEKA as an educational and research tool for data mining applications in the transportation safety domain. Its graphical interface, comprehensive algorithm library, and built-in evaluation framework make it accessible to practitioners without deep programming expertise, democratizing the use of machine learning for road safety analysis.

9.2 Future Scope

The Road Accident Analysis System developed in this project provides a solid foundation that can be extended and enhanced in several important directions in future work.

Real-World Dataset Integration: The current system uses a simulated dataset of 500 records. Future work should integrate the system with real accident datasets from national and state-level traffic databases, such as the Ministry of Road Transport and Highways (MoRTH) accident database in India. Larger, real-world datasets will provide more comprehensive training data and produce more reliable and generalizable models.

Additional Attributes: The current dataset includes six attributes. Future versions could incorporate additional attributes such as weather conditions, road surface condition, visibility, speed limit, presence of traffic signals, and driver alcohol status. These additional attributes are likely to improve classification accuracy and provide deeper insights into accident causation.

Deep Learning Models: Future work could explore the application of deep learning algorithms such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for accident severity prediction. These models can potentially capture complex non-linear relationships in the data that traditional machine learning algorithms may miss.

Web-Based Application: The current system operates as a desktop WEKA application. Future development could wrap the data mining capabilities in a user-friendly web-based interface, enabling real-time data entry and instant severity prediction for newly reported accidents.

Geospatial Analysis: Integrating the accident data with geographic information system (GIS) data would enable spatial analysis of accident patterns, identifying accident hotspots on road networks and supporting infrastructure improvement decisions.

Real-Time Prediction API: A RESTful API could be developed to expose the trained classification model as a service, enabling integration with traffic management systems, mobile applications, and emergency response platforms for real-time accident severity assessment.

CHAPTER 10: BIBLIOGRAPHY

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APPENDIX-A

Pseudo Code for Classification Algorithms Used in Road Accident Analysis System

A.1 Pseudo Code for J48 Decision Tree Algorithm

```
ALGORITHM: J48 Decision Tree (C4.5)
INPUT: Training dataset D with attributes A1, A2, ..., An and class C
OUTPUT: Decision Tree T

PROCEDURE BuildTree(D, A):
  IF all instances in D belong to same class Ci THEN
    RETURN Leaf node labeled Ci
  END IF

  IF A is empty THEN
    RETURN Leaf node labeled with majority class in D
  END IF

  // Select best splitting attribute
  FOR EACH attribute Aj in A DO
    GainRatio(Aj) = InfoGain(D, Aj) / SplitInfo(D, Aj)
    InfoGain(D, Aj) = Info(D) - InfoA(D, Aj)
    Info(D) = - SUM [p(Ci) * log2(p(Ci))]
    InfoA(D, Aj) = SUM [|Dv|/|D| * Info(Dv)]
  END FOR

  BestAttr = argmax(GainRatio(Aj))
  Create node N labeled with BestAttr

  FOR EACH value v of BestAttr DO
    Dv = subset of D where BestAttr = v
    IF Dv is empty THEN
      Add leaf node with majority class of D
    ELSE
      Add subtree BuildTree(Dv, A - {BestAttr})
    END IF
  END FOR

  Apply post-pruning using confidence factor CF = 0.25
  RETURN Node N
END PROCEDURE
```

A.2 Pseudo Code for Naive Bayes Classifier

```
ALGORITHM: Naive Bayes Classifier
INPUT: Training dataset D, Test instance x = {a1, a2, ..., an}
```

```
OUTPUT: Predicted class label C*

TRAINING PHASE:
  FOR EACH class value c in {Minor, Moderate, Severe} DO
    P(c) = Count(c) / |D|  // Prior probability
    FOR EACH attribute Aj DO
      FOR EACH value v of Aj DO
        P(Aj=v | c) = Count(Aj=v AND class=c) / Count(class=c)
        // Laplace smoothing to handle zero probabilities
        P(Aj=v | c) = (Count(Aj=v, c) + 1) / (Count(c) + |values(Aj)|)
      END FOR
    END FOR
  END FOR

CLASSIFICATION PHASE:
  FOR EACH class value c DO
    Posterior(c) = P(c) * PRODUCT [P(aj | c)] for each attribute
  END FOR
  C* = argmax_c [Posterior(c)]
  RETURN C*
```

A.3 Pseudo Code for Random Forest Algorithm

```
ALGORITHM: Random Forest
INPUT: Training dataset D, Number of trees T=100, Features per split m=2
OUTPUT: Ensemble classifier RF

TRAINING PHASE:
  FOR i = 1 TO T DO
    // Bootstrap sampling
    Di = BootstrapSample(D)  // Sample |D| instances with replacement

    // Build decision tree with random feature selection
    Ti = BuildRandomTree(Di, m):
      AT EACH NODE:
        F_sub = RandomSubset(All_Features, m)
        BestSplit = argmax_f_in_F_sub [InfoGain(f, node_data)]
        Split node on BestSplit
        Recurse on child nodes
      NO PRUNING applied

    Add Ti to ensemble RF
  END FOR

CLASSIFICATION PHASE:
  FOR test instance x DO
    Votes = {}
    FOR EACH tree Ti in RF DO
      Votes[Ti.classify(x)] += 1
    END FOR
    C* = argmax_c [Votes[c]]  // Majority vote
    RETURN C*
```

```
END FOR
```

The pseudo code presented in this appendix provides a formal algorithmic description of the three classification methods used in the Road Accident Analysis System. These descriptions serve as a reference for understanding the internal workings of the WEKA implementations and for replicating the algorithms in other programming environments.